

University of California Riverside
Electrical and Computer Engineering Department

Welcome to EE 116
Engineering
Electromagnetics

March 31, 2026

Today: Introductory Lecture

Outline

- A brief course overview
- Lecture scheduler
- Exams and preparation to exams
- Grading
- Questions?
- A little bit of history

Meeting Days/Time

- In-person, Tuesday and Thursday
- 02:00 PM - 03:20 PM
- 03/30/2026 - 06/05/2026
- Riverside Campus | Student Success Center | Room 335

The best way to communicate

- Please send an email:

akhitun@ece.ucr.edu

Please write EE116 in the subject line

Do not hesitate to send a second email if you do not have a response within 24 hours

Office Hours

Thursday: 12:00 PM - 1:00 PM

Room: WCH 349

Please send an email to: akhitun@ece.ucr.edu

Subject line: EE 116 Office Hours

Lecture Notes

- ❑ All lecture notes will be posted on iLearn.
- ❑ Pdf file with notes will be posted before the lecture.
- ❑ Link to the lecture recorded will be posted after the lecture.
- ❑ Solutions (Homework, Tests, Midterm) will be posted on iLearn as well.
- ❑ Please check iLearn for announcements and assignments.

Teaching Assistant

Mahmud, Quazi Sanjid

Email:

quazisanjid.mahmud@email.ucr.edu

Please send your questions regarding the assignment scores to Sanjid.

Electromagnetics is the fundamental physics of electrical engineering, and describes how electric charges and currents create electric and magnetic fields.

- Antennas
- Coaxial cable
- Design and characterization of common discrete passive components including resistors, capacitors, inductors, and diodes
- Fiber optics
- Generators
- Magnetic resonance imaging (MRI)
- Magnetic storage (of data)
- Microstrip transmission lines
- Motors
- Non-contact sensors
- Photonics
- Radio wave propagation
- Radio frequency electronics
- Signal integrity
- Transformers
- Waveguides

Text Books

M.F. Iskander, *Electromagnetic Fields and Waves*, Prentice Hall, 1992.

Note: editions of different years are OK for the use.

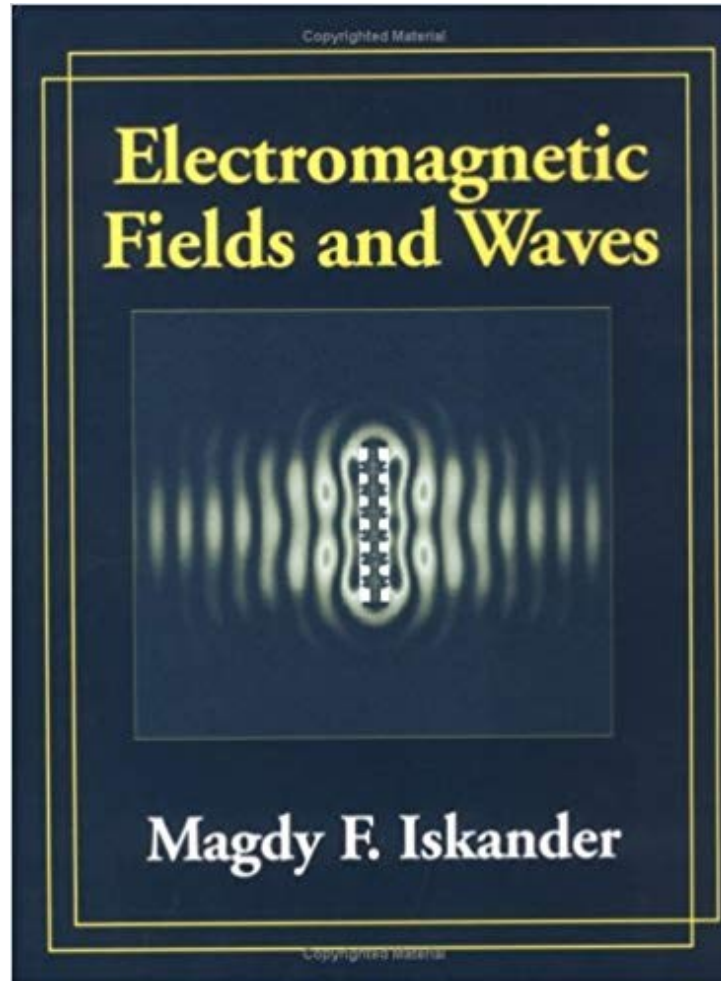
Additional Reading

(Ref1) John David Jackson, "Classical Electrodynamics", Wiley, 1998

(Ref2) David J. Griffiths, "Introduction to Electrodynamics", Addison-Wesley, 2012

EE 116

- Maxwell's equations, Integral and Differential form
- Solutions in Cartesian, Cylindrical, and Spherical coordinate systems
- Static Electric and Magnetic Fields
- Boundary conditions for E and H fields
- Transmission lines
- Introduction to antennas



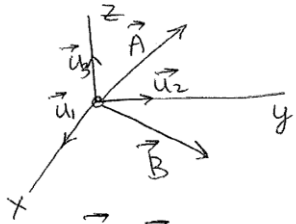
EE 117

- Maxwell's equations, Poynting's Theorem, uniqueness
- Plane wave propagation, reflection/transmission, and polarization
- Electromagnetic wave propagation in lossy, and dispersive media
- Metallic and dielectric wave guides
- Electromagnetic waves in fiber optics/photonics

Chapter 1: Vector Analysis and Maxwell's Equations In Integral Form

The vector $\vec{A}$ can be represented in terms of its components

$$\vec{A} = A_x \vec{a}_x + A_y \vec{a}_y + A_z \vec{a}_z$$



$$\vec{A} = A_1 \vec{u}_1 + A_2 \vec{u}_2 + A_3 \vec{u}_3$$

$$\vec{B} = B_1 \vec{u}_1 + B_2 \vec{u}_2 + B_3 \vec{u}_3$$

$$\vec{A} + \vec{B} = (A_1 + B_1) \vec{u}_1 + (A_2 + B_2) \vec{u}_2 + (A_3 + B_3) \vec{u}_3$$

Properties of unit vectors:

$$\vec{u}_1 \cdot \vec{u}_2 = 0$$

$$\vec{u}_1 \cdot \vec{u}_1 = 1$$

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_1 \vec{u}_1 + A_2 \vec{u}_2 + A_3 \vec{u}_3) \cdot (B_1 \vec{u}_1 + B_2 \vec{u}_2 + B_3 \vec{u}_3) = \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3 \end{aligned}$$

The cross product

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} =$$

$$= \vec{u}_1 \begin{vmatrix} A_2 & A_3 \\ B_2 & B_3 \end{vmatrix} - \vec{u}_2 \begin{vmatrix} A_1 & A_3 \\ B_1 & B_3 \end{vmatrix} + \vec{u}_3 \begin{vmatrix} A_1 & A_2 \\ B_1 & B_2 \end{vmatrix} =$$

$$= \vec{u}_1 (A_2 B_3 - A_3 B_2) - \vec{u}_2 (A_1 B_3 - A_3 B_1) + \vec{u}_3 (A_1 B_2 - A_2 B_1)$$

$$= \vec{u}_1 (A_2 B_3 - A_3 B_2) + \vec{u}_2 (A_3 B_1 - A_1 B_3) + \vec{u}_3 (A_1 B_2 - A_2 B_1)$$

This follows from the properties

$$\vec{u}_1 \times \vec{u}_2 = \vec{u}_3, \quad \vec{u}_2 \times \vec{u}_3 = \vec{u}_1, \quad \vec{u}_3 \times \vec{u}_1 = \vec{u}_2$$

$$\vec{u}_1 \times \vec{u}_1 = \vec{u}_2 \times \vec{u}_2 = \vec{u}_3 \times \vec{u}_3 = 0$$

Cartesian Coordinates

(x,y,z)

Differential length:

$$d\mathbf{l} = \hat{x}dx + \hat{y}dy + \hat{z}dz$$

Differential surface area:

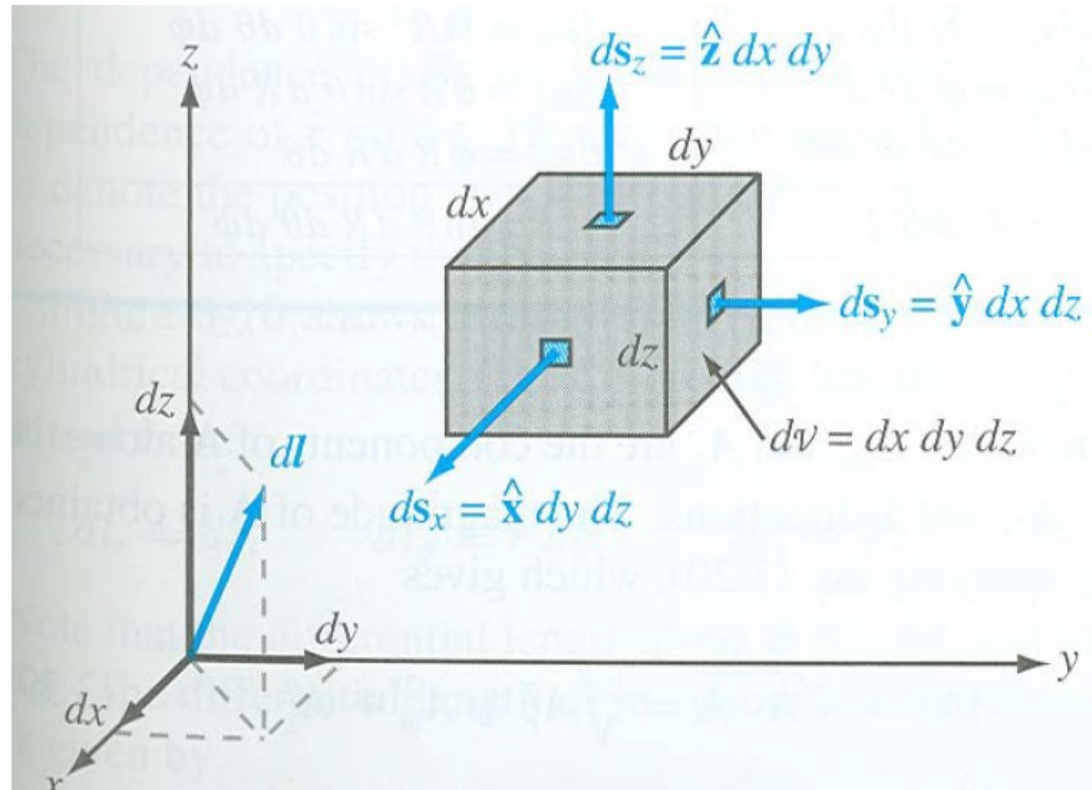
$$d\mathbf{s}_x = \hat{x}dydz$$

$$d\mathbf{s}_y = \hat{y}dxdz$$

$$d\mathbf{s}_z = \hat{z}dxdy$$

Differential volume:

$$dv = dxdydz$$



Chapters 1 and 2: Maxwell's Equations

INTEGRAL FORMS

$$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v dv$$

$$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$$

$$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s}$$

$$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s} + \frac{d}{dt} \int_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s}$$

DIFFERENTIAL FORMS

$$\nabla \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v \quad (4.1)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (4.2)$$

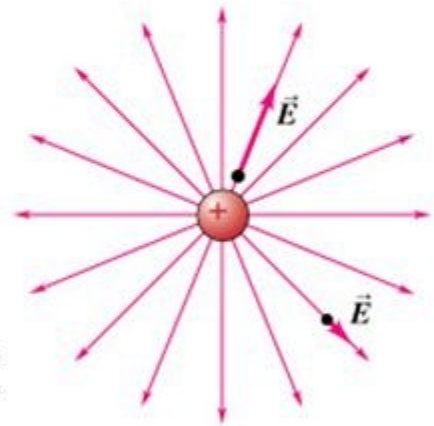
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (4.3)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \epsilon_o \epsilon_r \mathbf{E}}{\partial t} \quad (4.4)$$

Electric field equation for a point charge

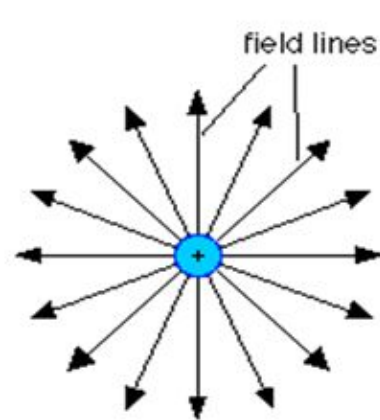
$$\vec{E} = \frac{\vec{F}_E}{q_0} \text{ becomes } E = k_c \frac{q}{r^2} \text{ for a point charge}$$

- E is electric field strength, units: N/C
- E is a VECTOR
- q_0 is a + test charge, units: C
- q is the charge producing the field, units: C
- r is the distance between charges, Units: m
- F_E is the Electric Force, units: N
- k_c is the coulomb constant, $8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$

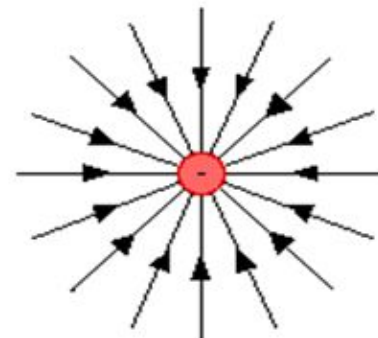


Electric Field Lines

- Electric fields are represented by electric field lines (*imaginary*) and show the direction of the force at that point.
- Electric field direction is from a positive charge to a negative charge.
- Electric field lines (also known as electric flux) do not touch each other.
- Electric field line represents the path a small positive charge would move in the electric field.



The electric field from an isolated positive charge

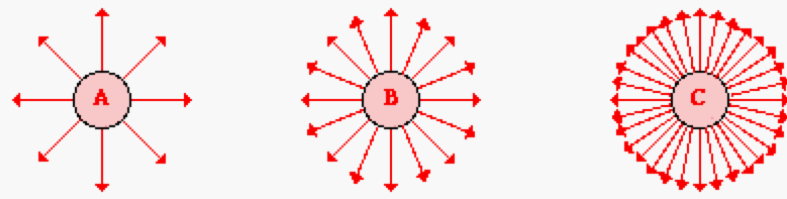


The electric field from an isolated negative charge

Electric Flux Density, $\mathbf{D}$

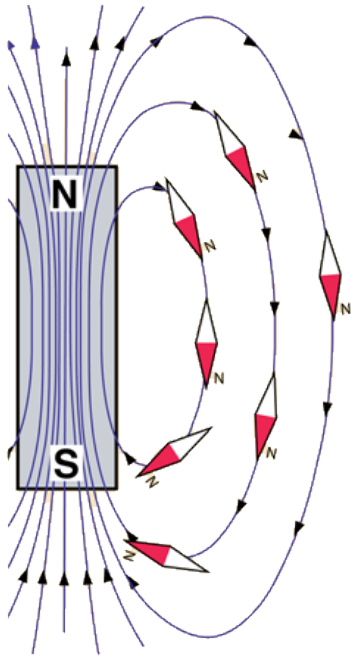
- Units: C/m^2
- Magnitude: Number of flux lines (coulombs) crossing a surface normal to the lines divided by the surface area.
- Direction: Direction of flux lines (same direction as $\mathbf{E}$).
- For a point charge: $\mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r$
- For a general charge distribution,

$$\mathbf{D} = \epsilon_0 \mathbf{E} = \int_{\text{vol}} \frac{\rho_v d\mathbf{v}}{4\pi R^2} \mathbf{a}_r$$

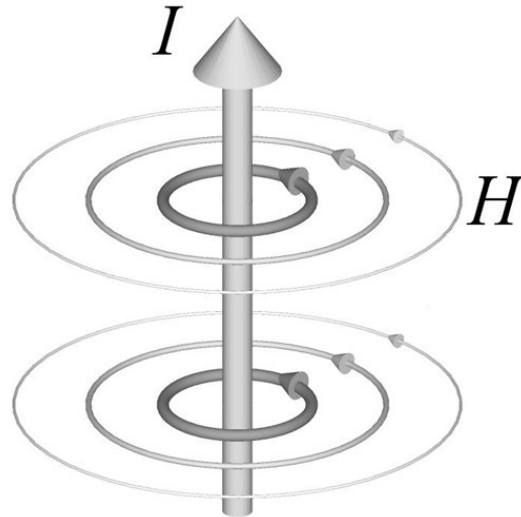


Magnetic Field Lines

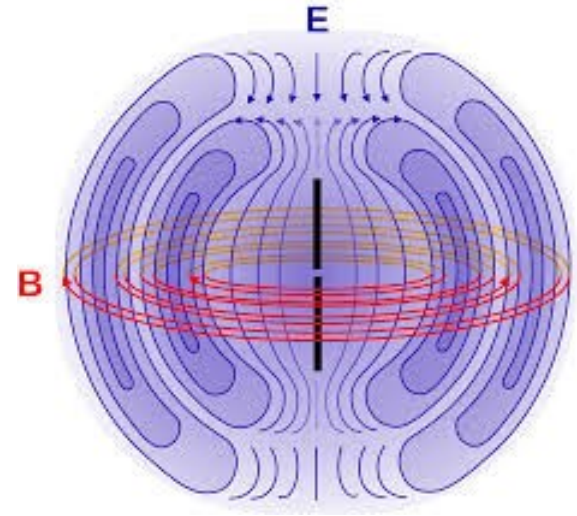
Permanent magnet



Electric current



Time-varying electric field



Mapping the magnetic field of an object is simple in principle. First, measure the strength and direction of the magnetic field at a large number of locations (or at every point in space). Then, mark each location with an arrow (called a vector) pointing in the direction of the local magnetic field with its magnitude proportional to the strength of the magnetic field.

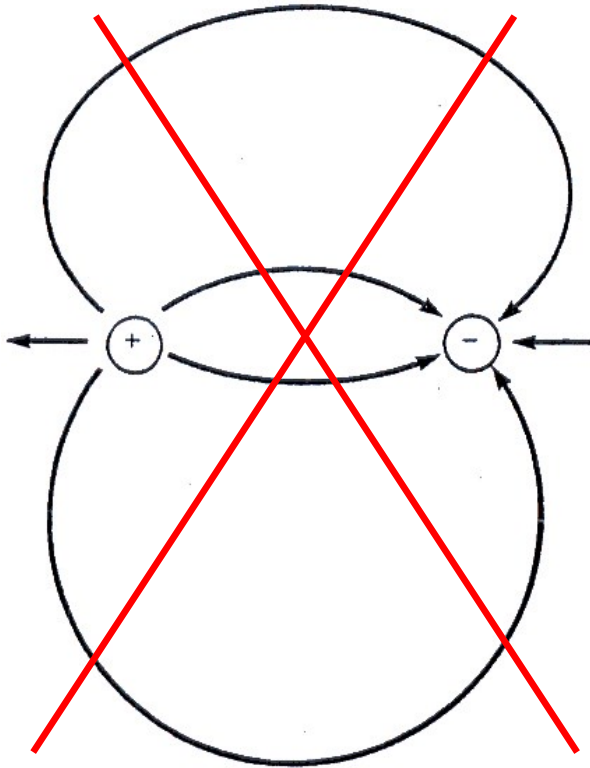
An alternative method to map the magnetic field is to 'connect' the arrows to form magnetic field lines. The direction of the magnetic field at any point is parallel to the direction of nearby field lines, and the local density of field lines can be made proportional to its strength. Magnetic field lines are like streamlines in fluid flow, in that they represent something continuous, and a different resolution would show more or fewer lines.

An advantage of using magnetic field lines as a representation is that many laws of magnetism (and electromagnetism) can be stated completely and concisely using simple concepts such as the 'number' of field lines through a surface. These concepts can be quickly 'translated' to their mathematical form. For example, the number of field lines through a given surface is the surface integral of the magnetic field. (Wikipedia)

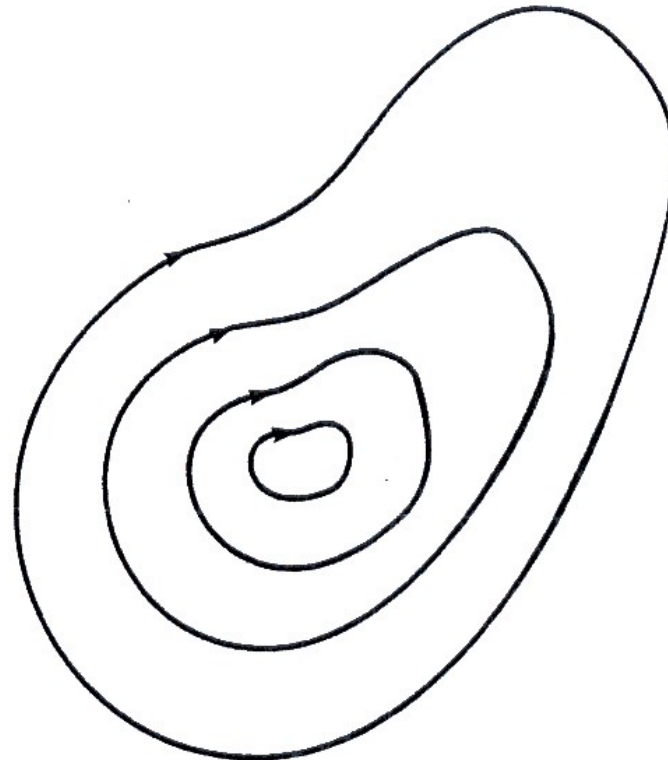
Different forms of Magnetic Field

Which of the graphs (left or right) shows magnetic field?

There are no magnetic charges!



Vortex electric field (e.g. produced by time varying magnetic field)



Chapter 3: Maxwell's Equations and Plane Wave Propagating in Materials

Waves in free space

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} \vec{\nabla} \times \vec{B} \quad (\text{3rd eq.})$$

$$\vec{\nabla} \times \vec{B} \text{ from 4th eq: } \vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\Rightarrow \vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

Vector Identity: $\vec{\nabla} \times \vec{\nabla} \times \vec{E} = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E}$

$$\vec{\nabla}^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \quad \text{Laplacian operator}$$

$$\vec{\nabla}^2 E_x = \frac{\partial^2 E_x}{\partial x^2} + \frac{\partial^2 E_x}{\partial y^2} + \frac{\partial^2 E_x}{\partial z^2}$$

$$\text{from 1st eq: } \vec{\nabla} \cdot \vec{E} = 0 \Rightarrow \vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\nabla^2 \vec{E}$$

$$-\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\boxed{\nabla^2 \vec{E} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0}$$

$$\boxed{\nabla^2 \vec{B} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} = 0}$$

— homogeneous vector wave equation

In Materials

$$\vec{E}(\mathbf{r}, t) = \vec{E}_0 e^{j(\omega t - \mathbf{k}_c \cdot \mathbf{r})} \equiv \vec{E}_0 e^{j\omega t} e^{-\gamma r} \quad \text{allow } \mathbf{k} \text{ be complex since } \epsilon \text{ is}$$

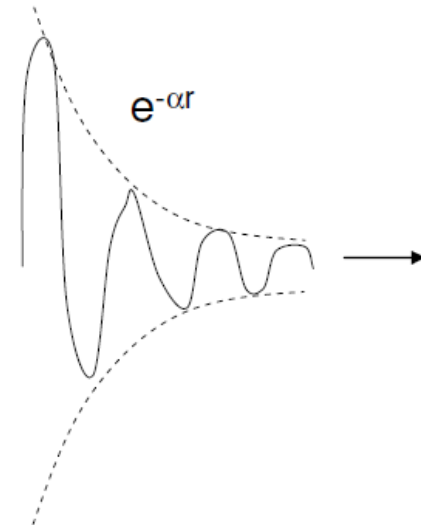
$$\gamma \equiv jk_c = j\omega\sqrt{\mu\epsilon_c} \equiv \alpha + j\beta$$

propagation constant

attenuation constant

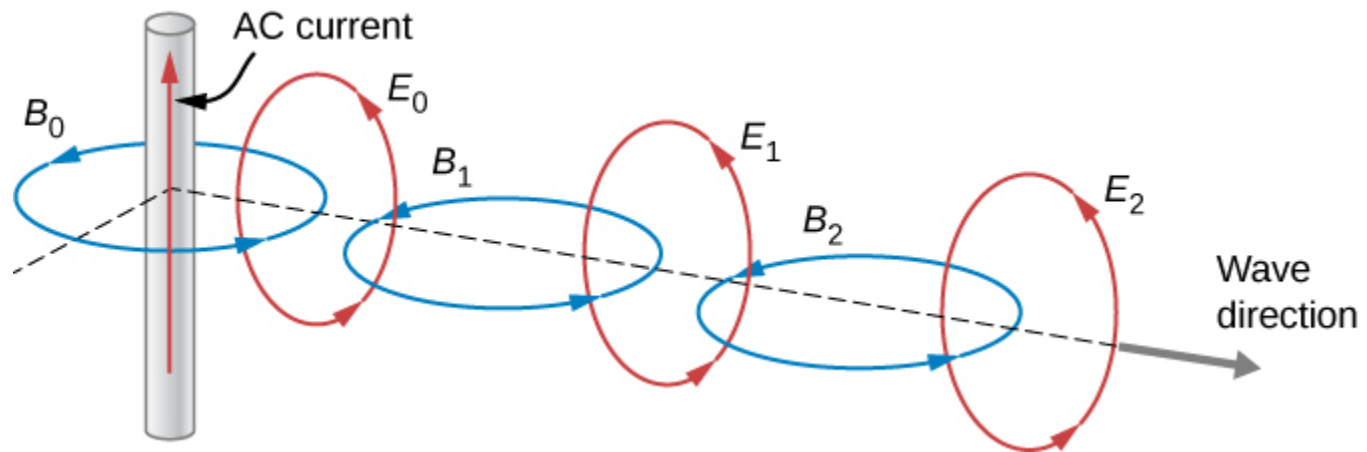
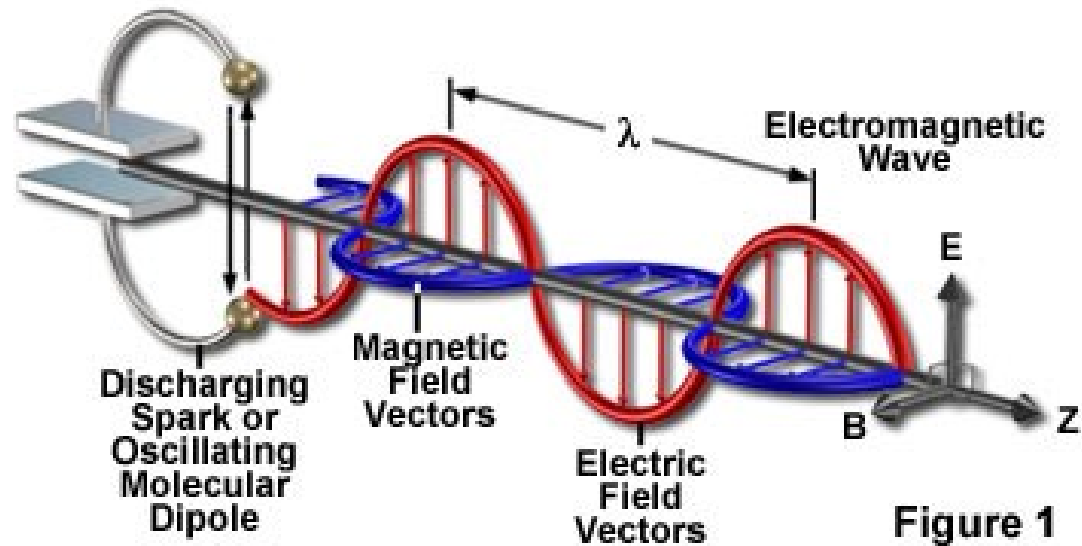
phase constant

$$\gamma = j\omega\sqrt{\mu\epsilon_c} = j\omega\sqrt{\mu\epsilon \left(1 - j\frac{\epsilon''}{\epsilon'}\right)} = j\omega\sqrt{\mu\epsilon \left(1 - j\frac{\sigma}{\omega\epsilon}\right)} = j\omega\sqrt{\mu\epsilon(1 - j\tan\delta)} = \alpha + j\beta$$



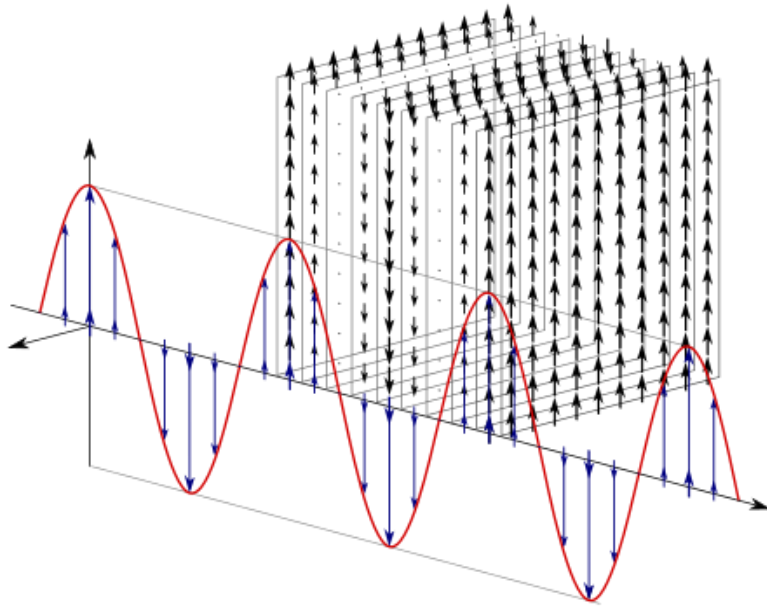
Propagating Electromagnetic Wave

Propagation of an Electromagnetic Wave

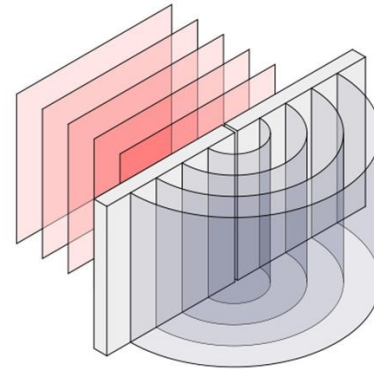


Plane, Cylindrical, and Spherical Electromagnetic Waves

Plane wave



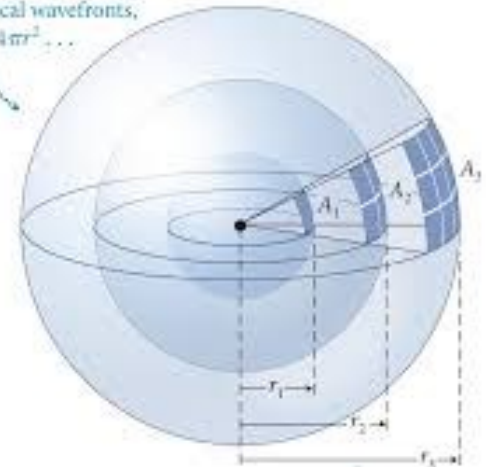
Cylindrical waves



$$\Psi = \left(\frac{A}{\sqrt{\rho}} \right) e^{i(k\rho \pm \omega t)}$$

Spherical Wave

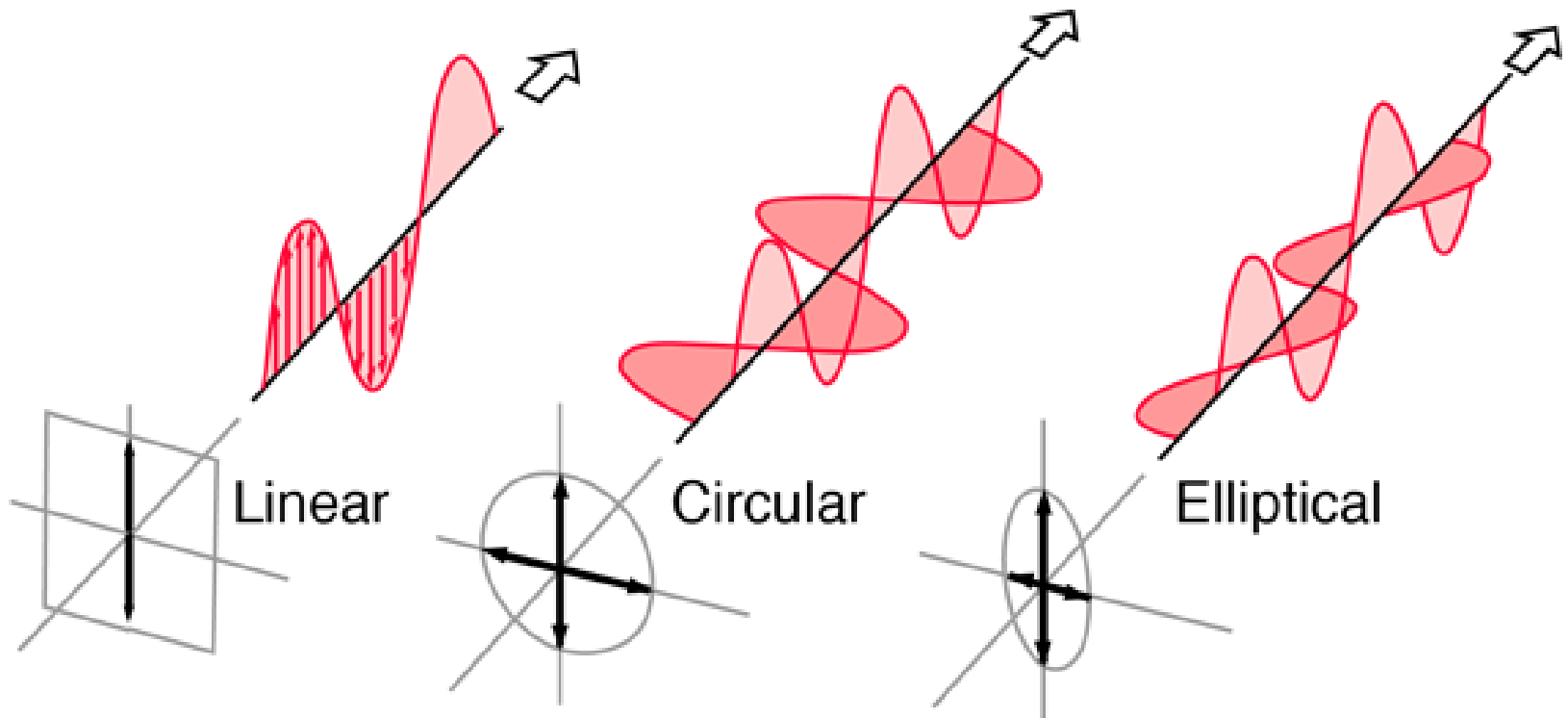
For spherical wavefronts,
area $A = 4\pi r^2 \dots$



... so energy per unit area decreases as $1/r^2$
(inverse-square relationship).

In physics, a wavefront is the locus of points characterized by propagation of positions of identical phase: propagation of a point in 1D, a curve in 2D or a surface in 3D. For an electromagnetic wave, the wavefront is represented as a surface of identical phase, and can be modified with conventional optics. For instance, a lens can change the shape of optical wavefronts from planar to spherical as the lens introduces a spatial phase variation across the beam shape.

Polarization of Plane Wave

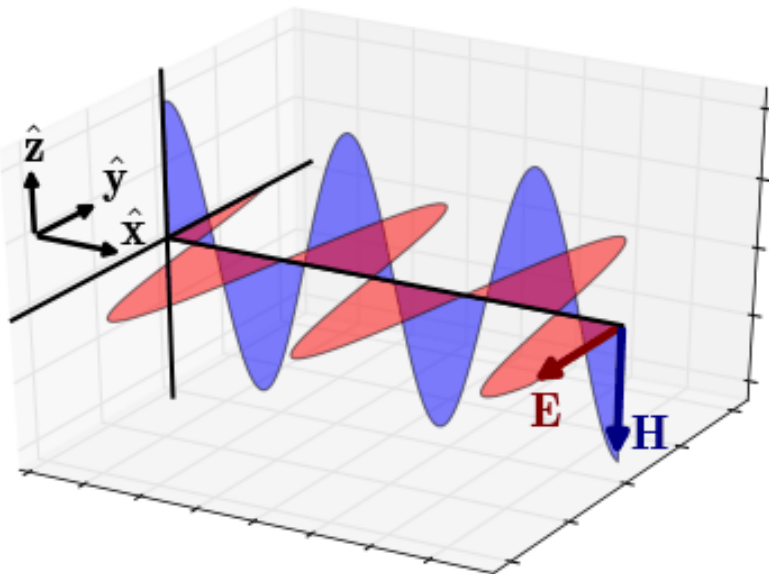


Skin Depth for Various Materials

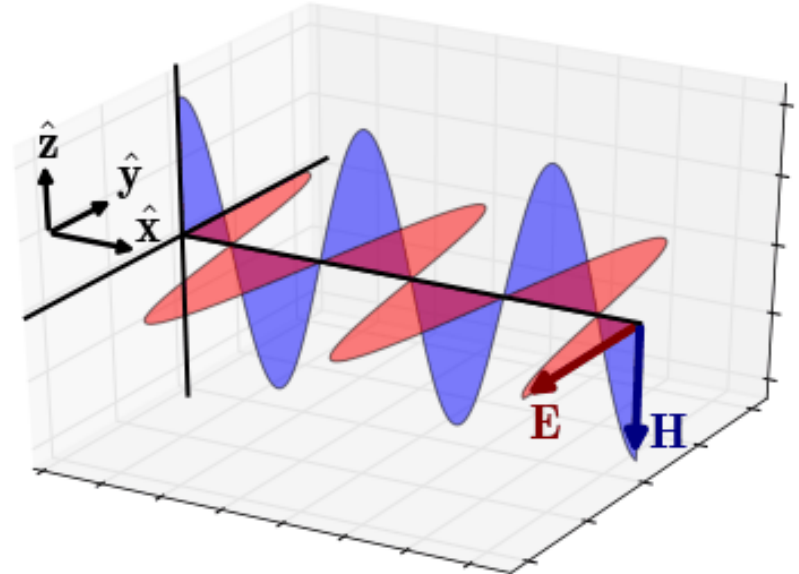
Type	σ	μ_r	ϵ_r	δ (1Hz)	δ (1kHz)	δ (1MHz)	δ (1GHz)
Vacuum	0 S/m	1	1	∞	∞	∞	∞
Sea Water	3.3 S/m	1	80	277 m	8.76 m	0.277 m	0.015 m
Igneous	10^{-4} S/m	1	5	50,300 m	1,590 m	121 m	119 m
Sedimentary (dry)	10^{-3} S/m	1	4	15,900 m	500 m	18 m	11 m
Sedimentary (wet)	10^{-2} S/m	1	25	5,000 m	160 m	5.4 m	2.6 m
Sulphide Skarn	10^2 S/m	1	5	50 m	1.6 m	0.05 m	0.002 m
Magnetite Skarn	10^2 S/m	2	5	36 m	1.1 m	0.04 m	0.001 m

Illustration

Quasi-static: $\frac{\sigma}{\omega\epsilon} \gg 1$

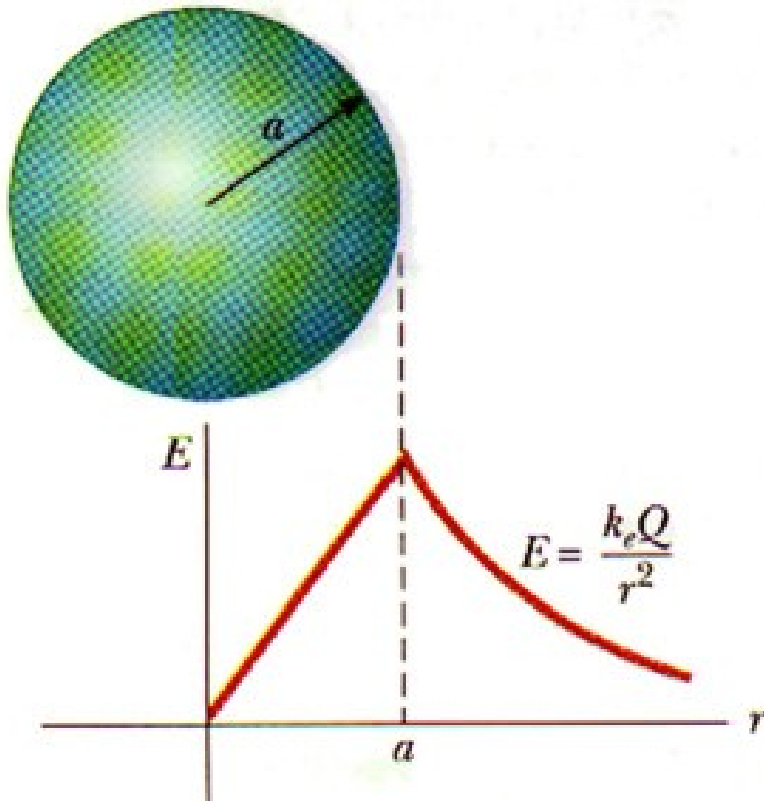


Wave regime: $\frac{\sigma}{\omega\epsilon} \ll 1$

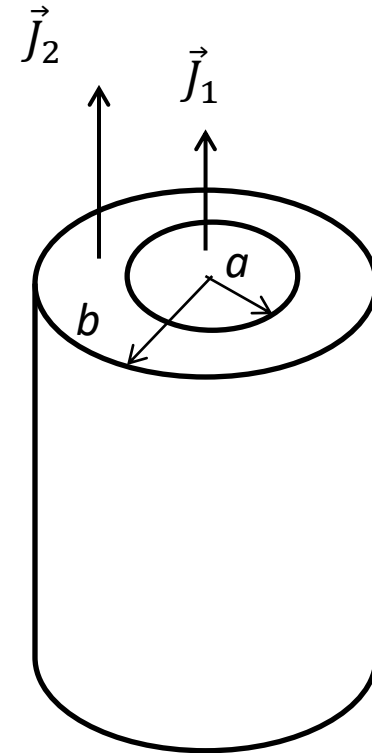


Chapter 4: Static Electric and Magnetic Fields

Electric Field produced by a charged sphere



Magnetic Field produced by a constant electric current



Current distribution in two coaxial cylinders



Chapter 7: Transmission Lines

Transmission Lines

Chapter 7
p 479 ①

Low frequency applications

Definition:

A transmission line is an electromagnetic guiding system consisting of two or more conductors embedded in a suitable dielectric.

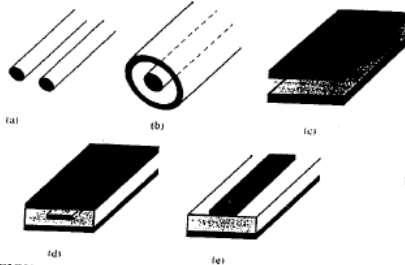


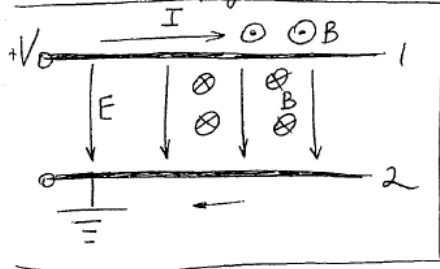
FIGURE 2.1. Different types of uniform transmission lines: (a) parallel two-wire; (b) coaxial; (c) parallel-plate; (d) stripline; (e) microstrip.

Uniform trans. lines: identical cross-section configuration for all positions along its length

TEM = principal mode

$f < 100 \text{ MHz}$

Field configuration: mode of propagation



TEM mode:

$$E_z = B_z = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$-\frac{\partial B}{\partial t} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & 0 \end{vmatrix} \Rightarrow \frac{\partial E_y}{\partial z} = \mu \frac{\partial H_x}{\partial t}$$

Reflection Coefficient

⑦

$$R_T/Z_0 = \frac{1 + \Gamma_T}{1 - \Gamma_T} \Rightarrow \boxed{\Gamma_T = \frac{R_T - Z_0}{R_T + Z_0}}$$

reflection coefficient at the resistive termination ↑

Matched Line $R_T = Z_0 \Rightarrow \Gamma_T = 0$

— load reflection coefficient is zero $\Rightarrow$
 $\Rightarrow$ no reflected voltages and currents

Open-circuited Line: $R_T = \infty \Rightarrow \Gamma_T = 1$

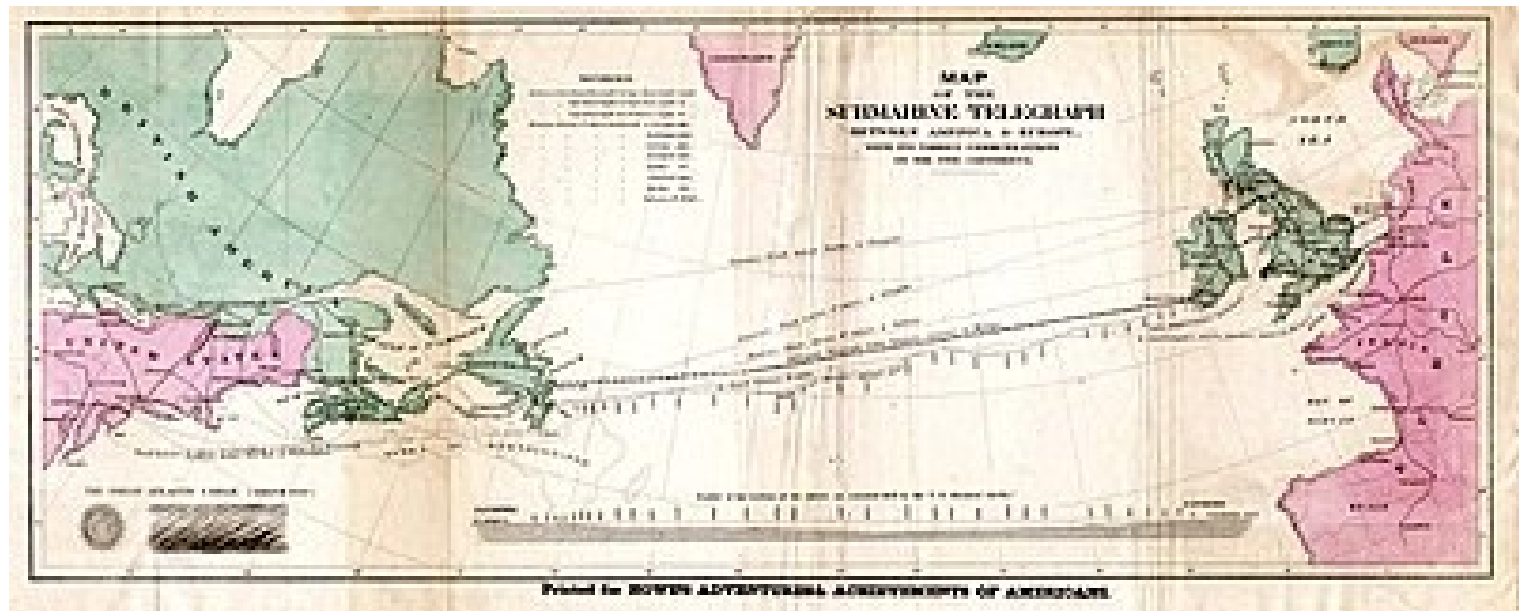
$\Rightarrow V_T^- = V_T^+ \Rightarrow$ the incident voltage is totally reflected by the load and the terminal voltage $V_T = 2V_T^+$ and current $I_T = 0$.

Short-circuited Line: $R_T = 0 \Rightarrow \Gamma_T = -1$

$\Rightarrow V_T^- = -V_T^+$ and $I_T^- = I_T^+$. The incident current is totally reflected by the load; $I_T = 2I_T^+$, $V_T = 0$

What happens in arbitrary terminations?

Transatlantic Telegraph Cable



Antennas. Introduction.

Antennas are devices that are used for transmitting and receiving electromagnetic energy, in the form of radio waves.

Antenna design:

- impedance matching to the feed transmission line
- best possible compromise between various constraints imposed by the desired radiation pattern.

Antennas

"wire antennas"

- straight wire
- loop
- helix



TV, radio stations,
radio receivers, etc.

"aperture antennas"

- horn 
- "lens antennas"
- "reflector antennas"

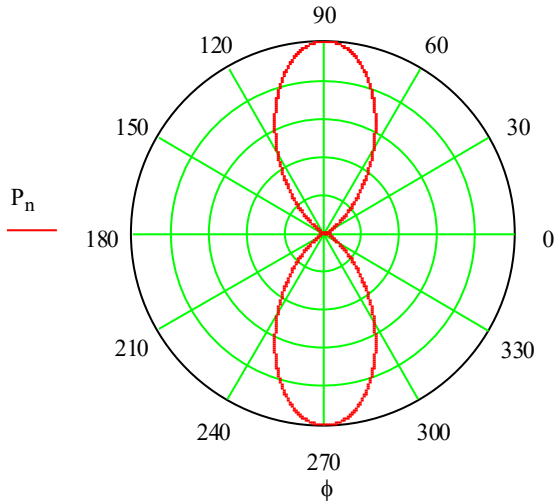


satellite communications



Chapter 9: Antennas

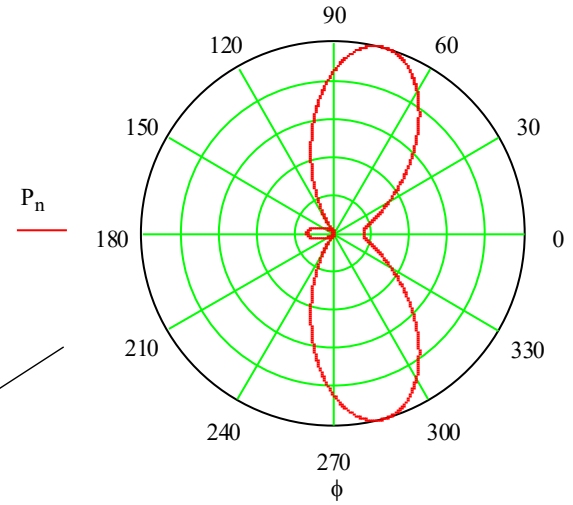
Power pattern of 2 isotropic sources



$$d = 1 \frac{\lambda}{2}$$

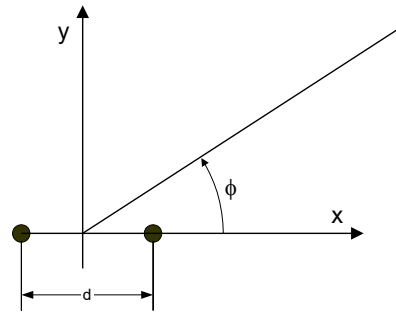
$$\delta = 0 \text{ deg}$$

Power pattern of 2 isotropic sources

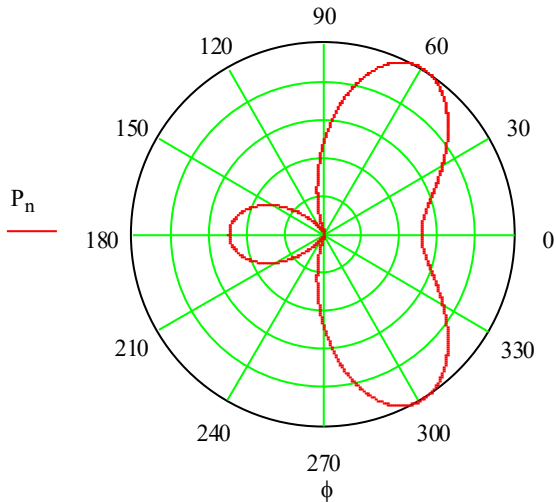


$$d = 1 \frac{\lambda}{2}$$

$$\delta = -45 \text{ deg}$$



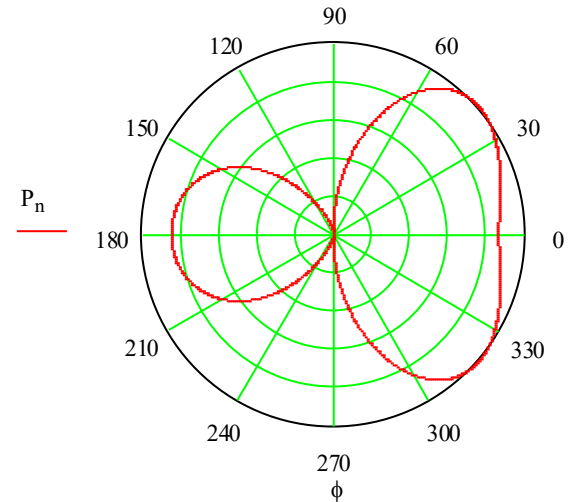
Power pattern of 2 isotropic sources



$$d = 1 \frac{\lambda}{2}$$

$$\delta = -90 \text{ deg}$$

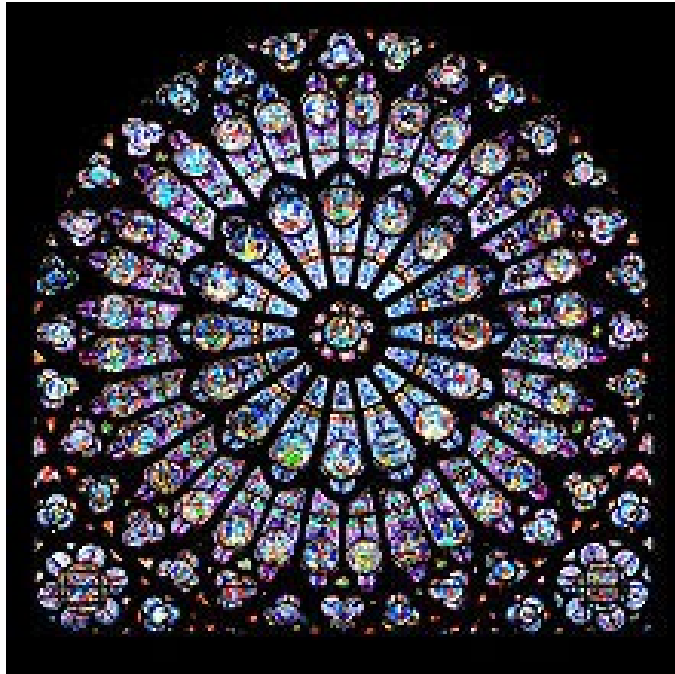
Power pattern of 2 isotropic sources



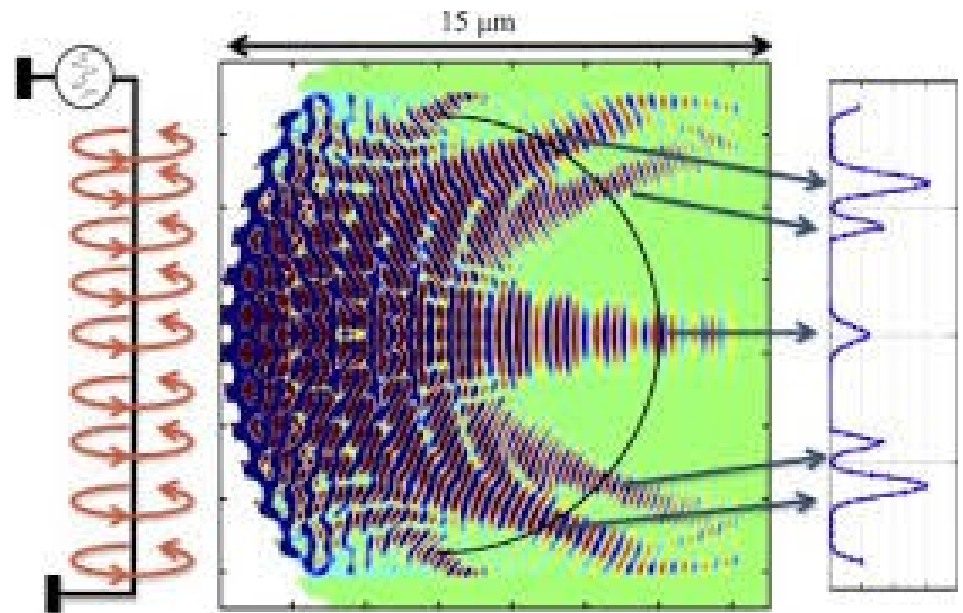
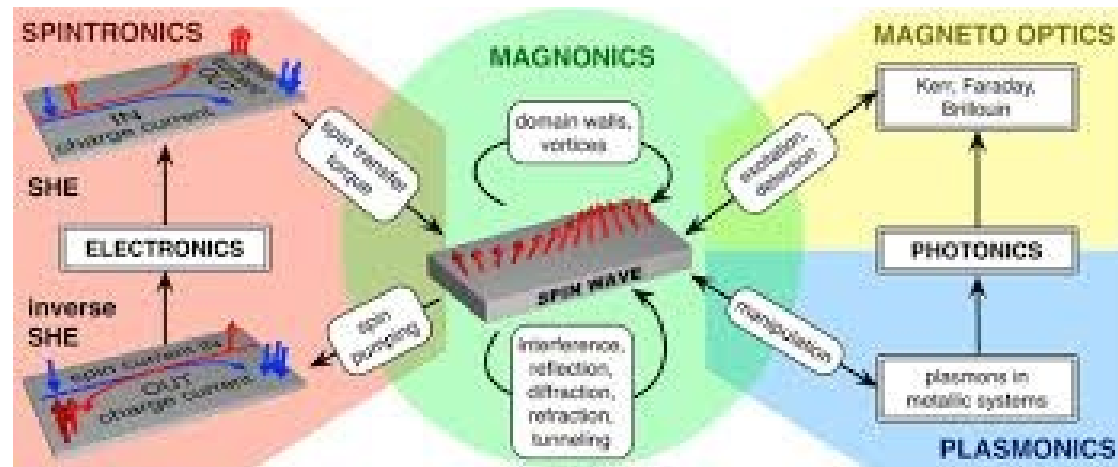
$$d = 1 \frac{\lambda}{2}$$

$$\delta = -135 \text{ deg}$$

Current advances in Electromagnetics



Gothic stained glass rose window of Notre-Dame de Paris. The colors were achieved by colloids of gold nano-particles.



Lecture Schedule

Week #	Lectures
#1	Introduction, Vector Calculus Chapter: 1
#2	Maxwell's Equations in Integral Form Chapters: 1.7
#3	Maxwell's Equations in Differential Form Chapter: 2
#4	Electromagnetic Waves Chapter: 2
#5	Dielectric and Magnetic Materials Chapters: 2,3. Preparation to the Midterm
#6	(Midterm) Maxwell's Equations in Material Region
#7	Static Electric and Magnetic Fields Chapter: 4
#8	Transmission lines Chapter: 7
#9	Transmission Lines, Introduction to antennas Chapter: 7
#10	Current advances in Electromagnetics (Online materials). Preparation to the Final Exam

Questions during the lecture

I may or may not answer questions during the lecture. All questions will be answered at the beginning of the next lecture.

Please check iLearn for announcements and assignments.

Grading

Examination and Grading:

Homework	10%
Midterm	40%
Final	50%
Total	100%

The Midterm and The Final exams are closed-book exams. However, you are allowed to have a one-page (A4) notes.

No-Fail tests are closed-book tests. 10 points (max) to the exam score.

A	90-100%	B	80-89%	C	70-79%
D	60-69%	F	< 60%		

+ and - are used in grading: i.e., 80-83 (B-); 83.1-87 (B); 87.1-89.9 (B+)

No-Fail Tests

Note: This is **NOT** an exam. It is a **no-fail test**. This test is intended to check students' understanding of lecture materials and homework.

You may earn extra credits in addition to your exam score.

Course Timeline

03/30/2026 - 06/05/2026

2:00 PM - 3:20 PM

APRIL 2026

SUN	MON	TUE	WED	THU	FRI	SAT
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	1	2

www.GrabCalendar.com

MAY 2026

SUN	MON	TUE	WED	THU	FRI	SAT
6	27	28	29	30	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

www.GrabCalendar.com

 No-fail test

 Midterm Exam

 Summary and Preparation for the Exam

Health or Family Emergency

All students will be accommodated in the case of health or family emergencies.

One needs to provide proof (e.g., a note from the doctor with the dates specified).

Questions?

Instructor: Dr. Alexander Khitun

Education and Professional Experience:

April 2011 - present	Research Engineer and Adjunct Professor, ECE, UC-Riverside
Jan 1999 - April 2011	Research Engineer, UCLA
Nov 1991 - Jan 1999	Research Fellow, Fiber Optics Research Center (FORC), General Physics Institute of the Russian Academy of Sciences
June 1995	Ph.D. Moscow Institute of Physics and Technology (MIPT), Russia
June 1991	M.S. Moscow Institute of Physics and Technology (MIPT), Russia

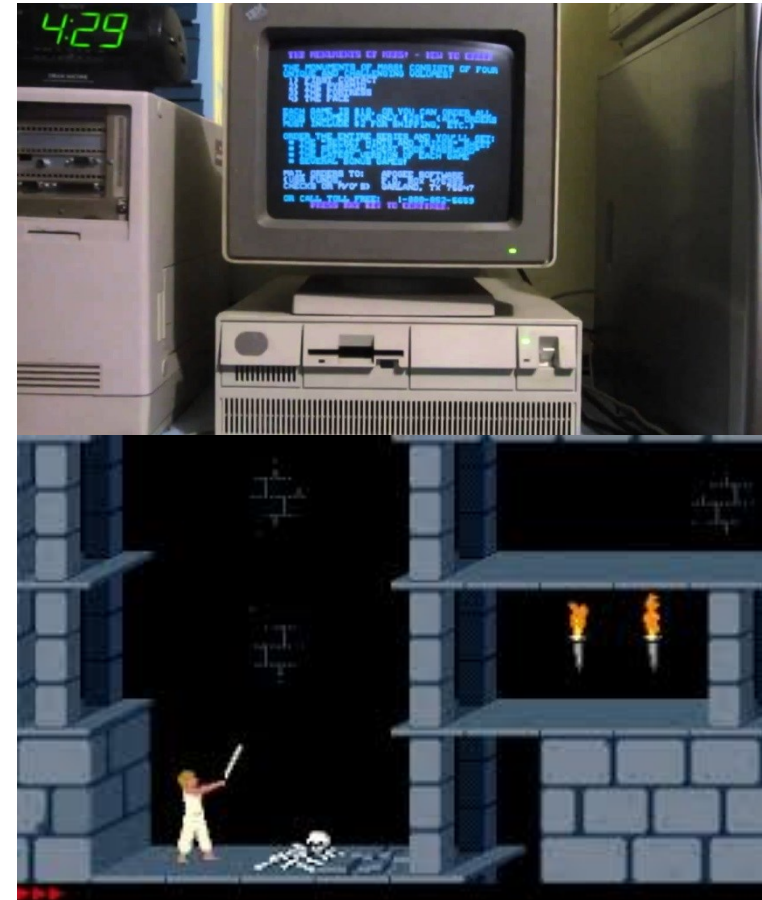
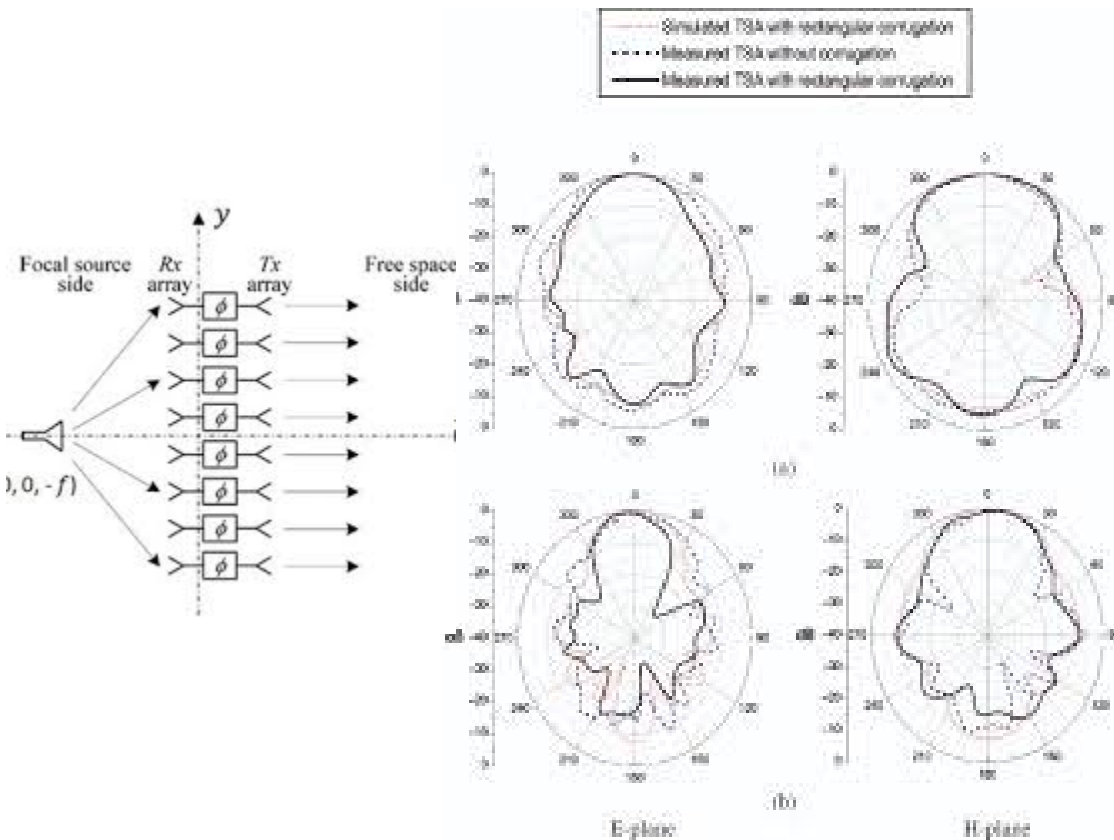
Co-authored 7 book chapters, more than 100 referred journal publications and symposia/conference proceedings, 8 US patents.

H-index: 39

Courses offered:

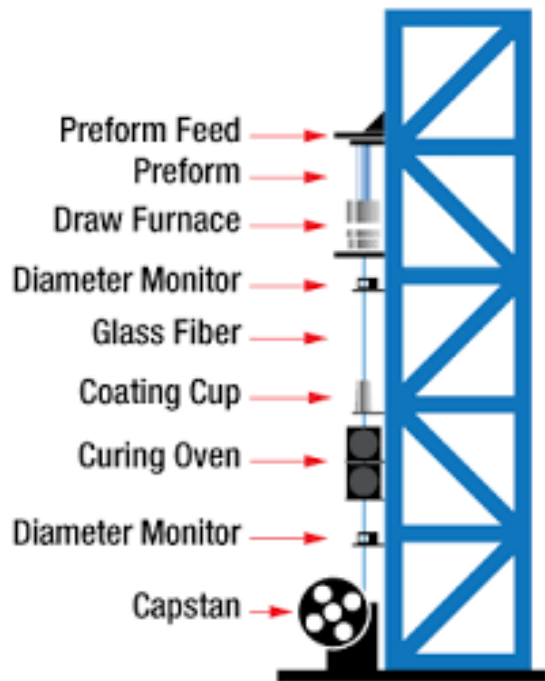
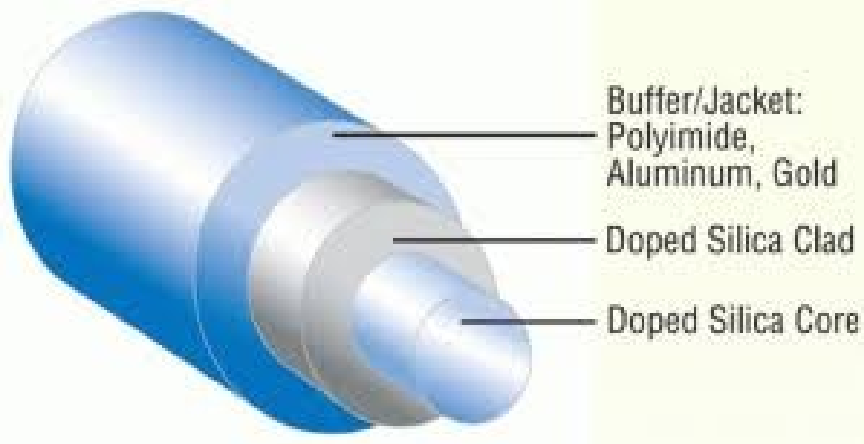
EE 01B Engineering Circuit Analysis II
EE 003 Electronics, Smartphones and Mobile Internet
EE116 Electromagnetics I
EE117 Electromagnetics II
EE133 Solid State Electronics
EE162 Introduction to Nanoelectronics
EE 203 Solid State Devices

Modeling Radiation Pattern from Antenna Arrays



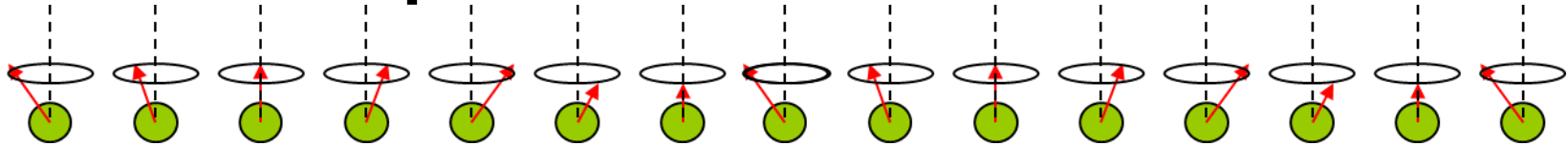
From Wiki: The Intel 80286 (and often called Intel 286) is a 16-bit microprocessor that was introduced on February 1, 1982. It was the first 8086 based CPU with separate, non-multiplexed address and data buses and also the first with memory management and wide protection abilities. The 80286 used approximately 134,000 transistors in its original nMOS (HMOS) incarnation and, just like the contemporary 80186, it could correctly execute most software written for the earlier Intel 8086 and 8088 processors.

Metal Coated Optical Fibers

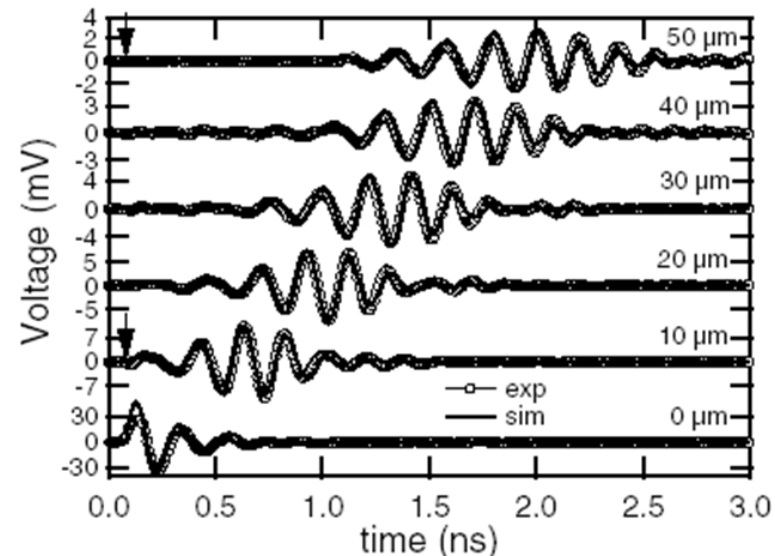
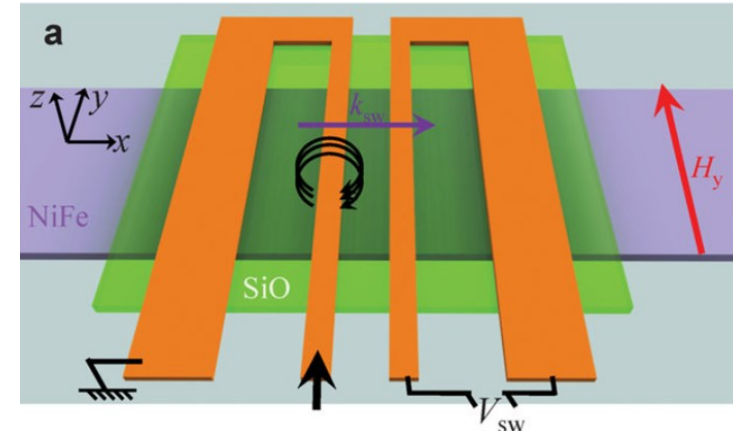


Fiber-Optic Guided Missile FOG-M

Spin Wave Devices

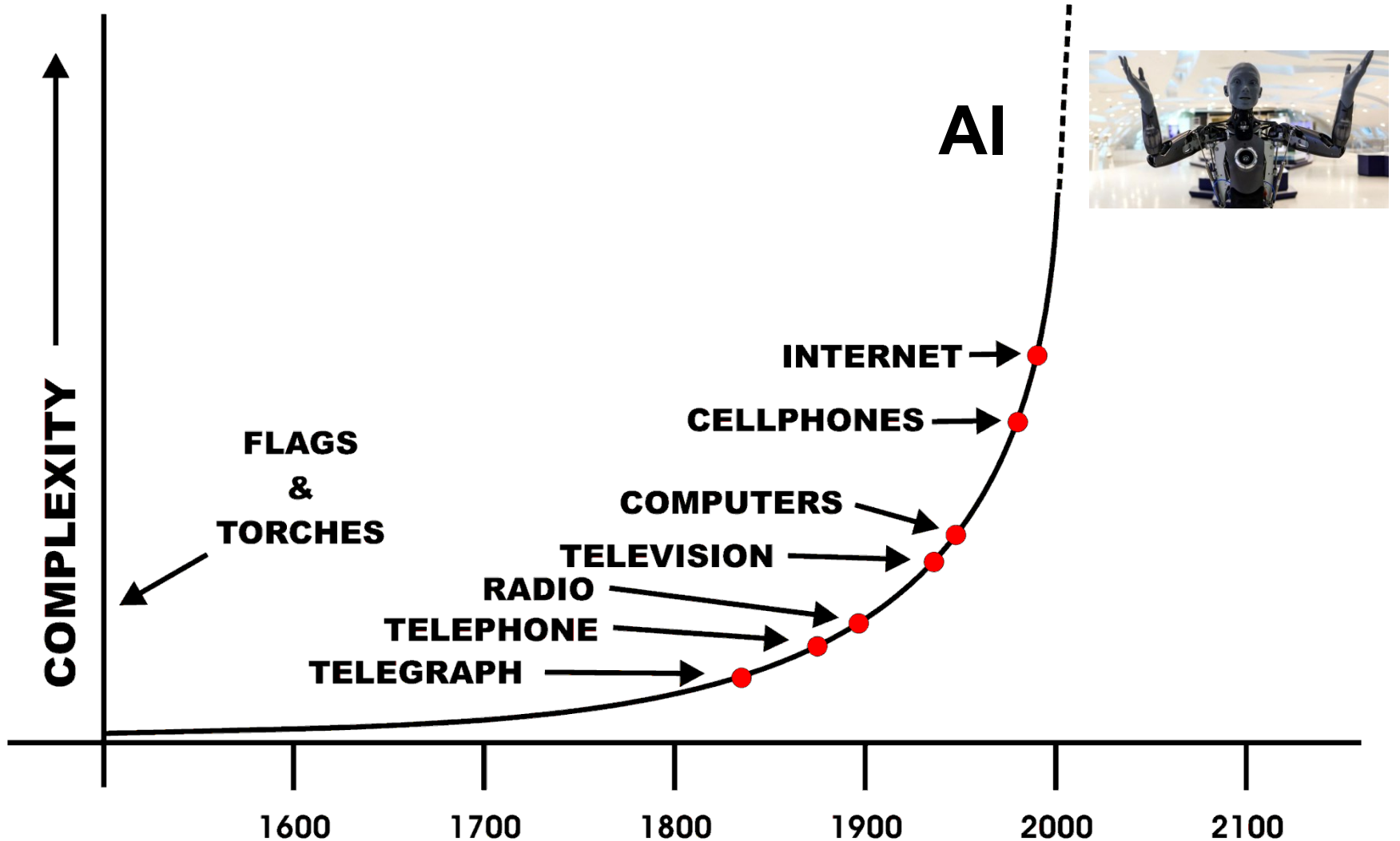


- ✓ Spin wave - collective oscillation of spins in magnetic lattice
- ✓ Sub-micrometer wavelength
- ✓ Frequency range (from GHz to THz)
- ✓ Group velocity $\sim 10^6$ cm/s
- ✓ Spin wave propagation does not involve electric current
- ✓ Propagating spin wave produces inductive voltage $V = -d\Phi/dt$
- ✓ Dispersion depends on the strength and direction of the magnetic field
- ✓ Interacts with magnetic objects via dipole-dipole / exchange interaction
- ✓ Coherence length at RT:
 - $\sim$ tens of microns in metals (e.g. NiFe)
 - $\sim$ 1cm in non-conducting ferrites (e.g. $Y_3Fe_2(FeO_4)_3$)



Covington, M., T.M. Crawford, and G.J. Parker, Time-resolved measurement of propagating spin waves in ferromagnetic thin films. Physical Review Letters, 2002. 89(23): p. 237202-1-4.

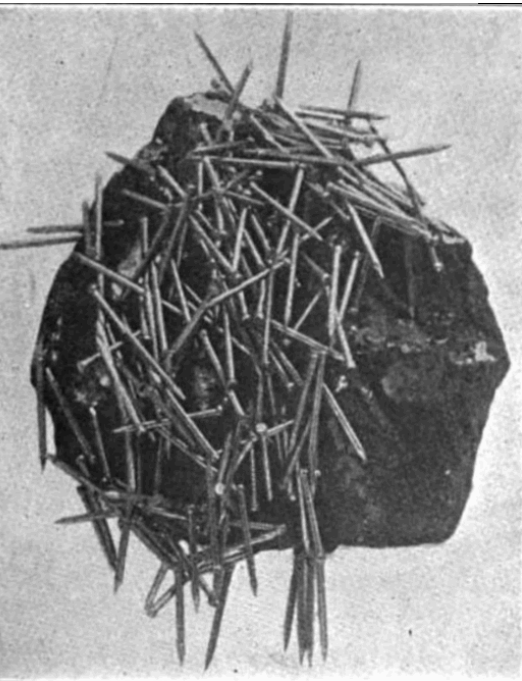
Timeline of Inventions



Electromagnetics in Ancient Time

600 BCE: Sparking Amber in Ancient Greece

The earliest writings about electromagnetism were in 600 BCE, when the ancient Greek philosopher, mathematician and scientist Thales of Miletus described his experiments rubbing animal fur on various substances such as amber. Thales discovered that amber rubbed with fur attracts bits of dust and hairs that create static electricity, and if he rubbed the amber for long enough, he could even get an electric spark to jump..



221–206 BCE: Chinese Lodestone Compass

The magnetic compass is an ancient Chinese invention, likely first made in China during the Qin dynasty, from 221 to 206 BCE. The compass used a lodestone, a magnetic oxide, to indicate true north. The underlying concept may not have been understood, but the ability of the compass to point true north was clear.



Model of a Han Dynasty (206 BC–220 AD) south-indicating ladle or sinan. It is theorized that the south-pointing spoons of the Han dynasty were magnetized lodestones.

Magnetism and Electricity

In 1600 **Gilbert** published the “*De Magnete, Magneticisque Corporibus, et de Magno Magnete Tellure*” (On the Magnet and Magnetic Bodies, and on That Great Magnet the Earth).

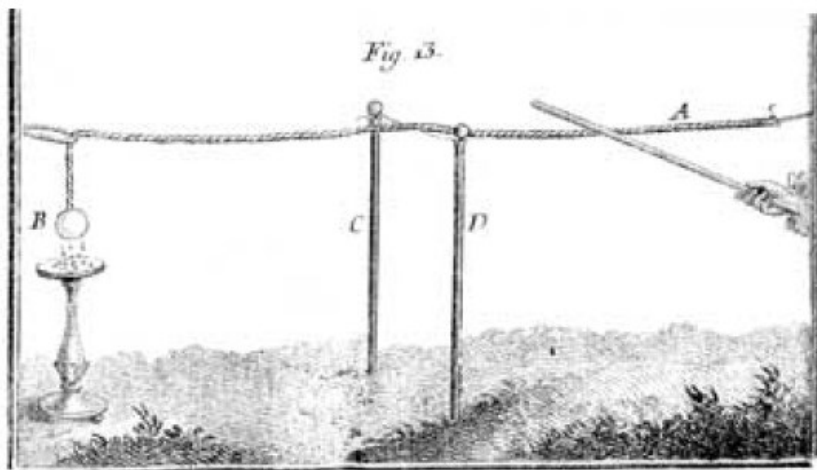
- ♦ The Earth is magnetic (Μαγνησία, Magnesia ad Sipylum) and this is why a compass points north.
- ♦ In a quite large class of bodies (glass, sulphur, ...) the friction induces the same effect observed in the amber (Ηλεκτρον, Elektron). Gilbert gave to it the name “*electricus*”.
- ♦ Gilbert remarked many distinction between the magnetic and the electric forces.
For instance, the lodestone requires no stimulus of friction such as it is needed to stir glass and sulphur into electrical activity.



William Gilbert
(England, 1540-1603)

Electric Current

In 1729 **Gray** discovered the “passage of electricity” through metals.



Stephen Gray
(England, 1666-1736)

Vitreous and Resinous Electricity

In 1733 **Du Fay** discover the two kinds of electricity: vitreous and resinous (for instance, sulfur is vitreous and amber is resinous).

We say nowadays positive and negative electricity, terms coined independently by **Watson** (1746) and **Franklin**.



Charles François
de Cisternay Du Fay
(France, 1698–1739)



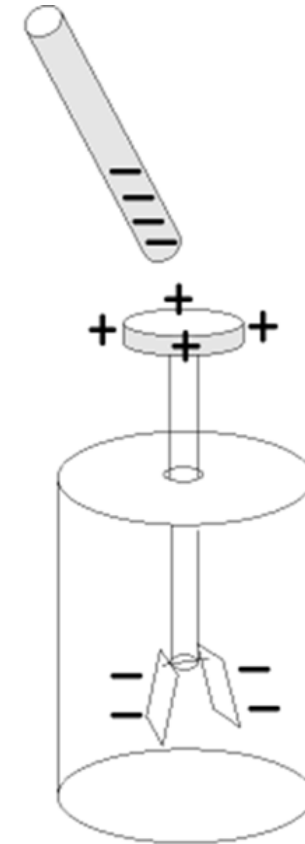
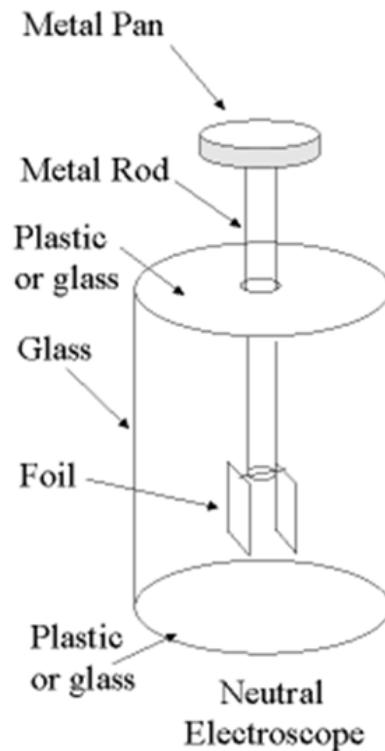
William Watson
(England, 1715 –1787)



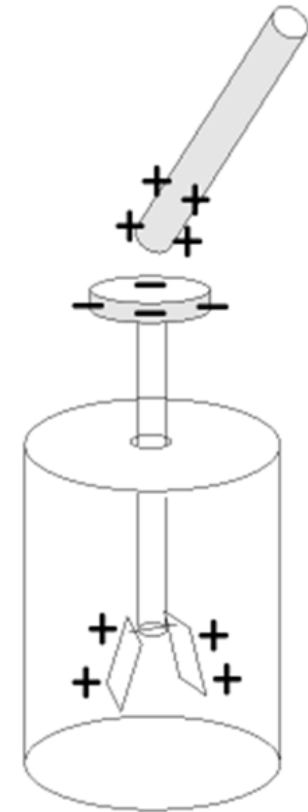
Benjamin Franklin
(United Staes, 1706-1790)

Independently, **Franklin** (1747) and **Watson** (1746) stated the principle of conservation of the quantity of electrical charge.

Example of Electric Charge Repulsion



Electroscope approached
with a negative rod
Like charge foils repel



Electroscope approached
with a positive rod
Like charge foils repel

Franklin's Kite Experiments

Franklin wanted to demonstrate the electrical nature of lightning. He had a simple kite made with a large silk handkerchief, a hemp string, and a silk string. He also had a house key, a Leyden jar (a device that could store an electrical charge for later use), and a sharp length of wire. His son William assisted him.

After his successful demonstration, Franklin continued his work with electricity, going on to perfect his lightning rod invention. In 1753, he received the prestigious Copley Medal from the Royal Society, in recognition of his “curious experiments and observations on electricity.”

A Leyden jar is an antique electrical component which stores a high-voltage electric charge (from an external source) between electrical conductors on the inside and outside of a glass jar.



Photograph of a painting depicting Franklin's famous kite and key experiment. Charles E. Mills (American, 1856-1956) | circa 1911 | Library of Congress, Washington, D.C.



It is a very dangerous experiment!



Georg Wilhelm Richmann
(1711 – 1753)



The wire is not grounded .The kite experiment resulted in a ball lightning



“And at that moment a bluish fireball the size of a large apple appeared in the air. He flew over to Richmann's head. There was a strong explosion. The luminous ball that killed an outstanding scientist was nothing more than ball lightning”.

“If you put your head under the wire, you will feel a stab. Also, when you put your shoulder to the wire, then you feel the stabbing through the dress. When the hammer is applied to the forehead and teeth, and the other end to the wire, you will feel a considerable illness. Small animals feel a greater disease than large ones.”

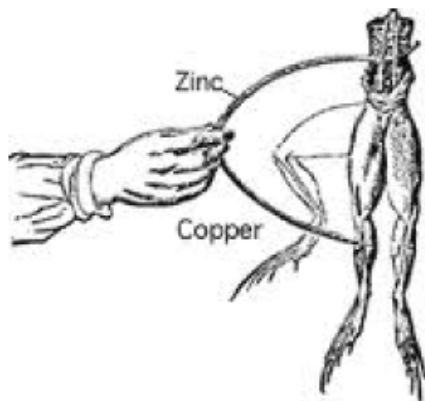
In another experiment, it was observed: “A head covered with hair, without a bald spot, when approaching an electrified iron wire, feels painful clicks.”

“If a rooster is electrified on an iron plate, then, in the case of a hand touching the ends of his legs, a hissing blue fire emanates.”

Electric Currents Chemically Generated

In 1780 **Galvani** discovered the possibility to generate chemically electric currents.

The metals used to mount and dissect the leg of frogs induced electric currents in their nerves.

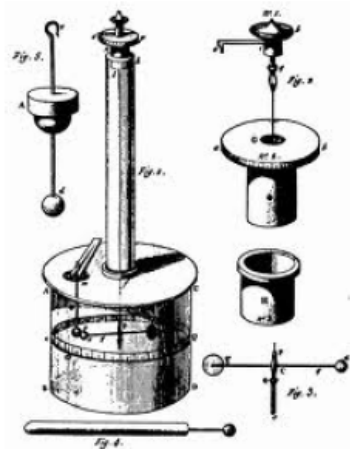


Luigi Galvani
(Italy, 1737-1798)

He gave rise to an interest in electricity created by the contacts between different materials.

Coulomb's Laws

In 1785 **Coulomb** formulated the law governing the force acting between **electric charges** and the law governing the force acting between **magnetic poles**.



Charles Augustin
de Coulomb
(France, 1736-1806)

- ◆ The **electricity** is composed of two electric fluids that can be separated: the parts of the same fluid repelling each other according to the inverse square of the distance, and attracting the parts of the other fluid according to the same inversion square law.
- ◆ The **magnetism** is also composed of two fluids, the *austral* and *boreal*, permanently imprisoned within the molecules of magnetic bodies, which obey the same inverse square law.

Direct Action at Distance *versus* Aether

“Until the seventeenth century the only influence which was known to be capable of passing from star to star was that of light. **Newton** added to this the **force of gravity**; ...” (Wittaker)

Newton attempted to account for gravitation by differences of pressure in an *aether*, but he did not publish his theory because he was not able from experiments and observations to give a satisfactory account of this medium.

The only *aether* that survived is that invented by **Huygens** (1690): the medium that penetrates all matter and is present even in all so-called vacuum, in which the propagation of light takes place.



Isaac Newton
(England, 1643 - 1727)

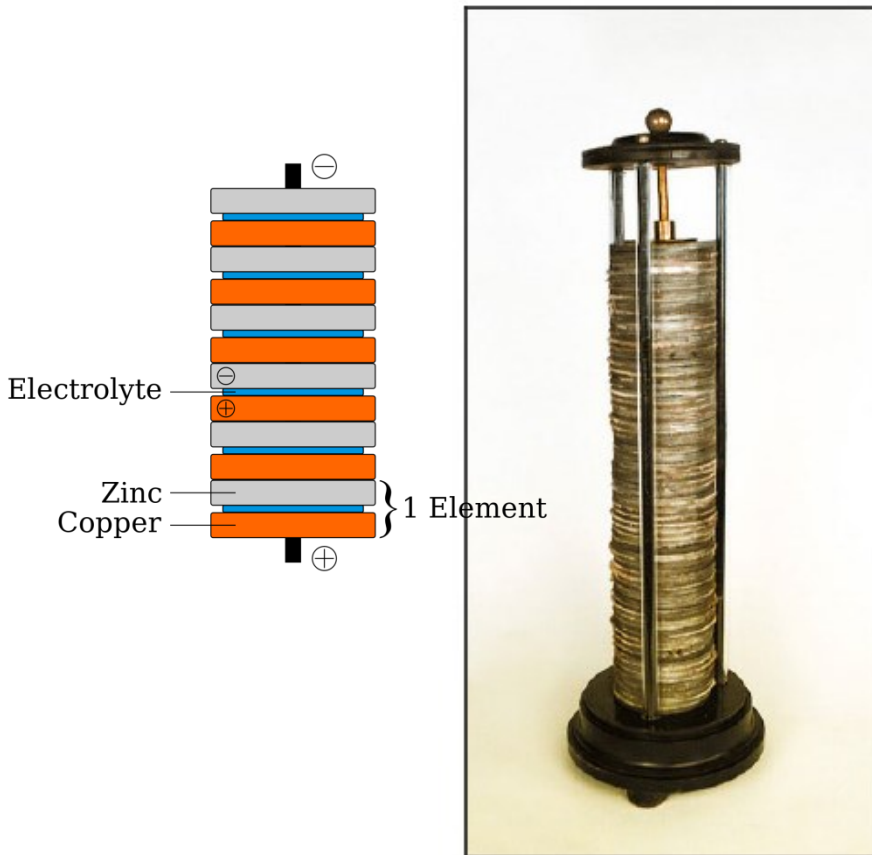


Christian Huygens
(Holland, 1629-1695)

After **Coulomb** it was recognized that the power of communicating directly across vacuous regions was possessed also by the **electric** and **magnetic interactions**.

Volta's Pila

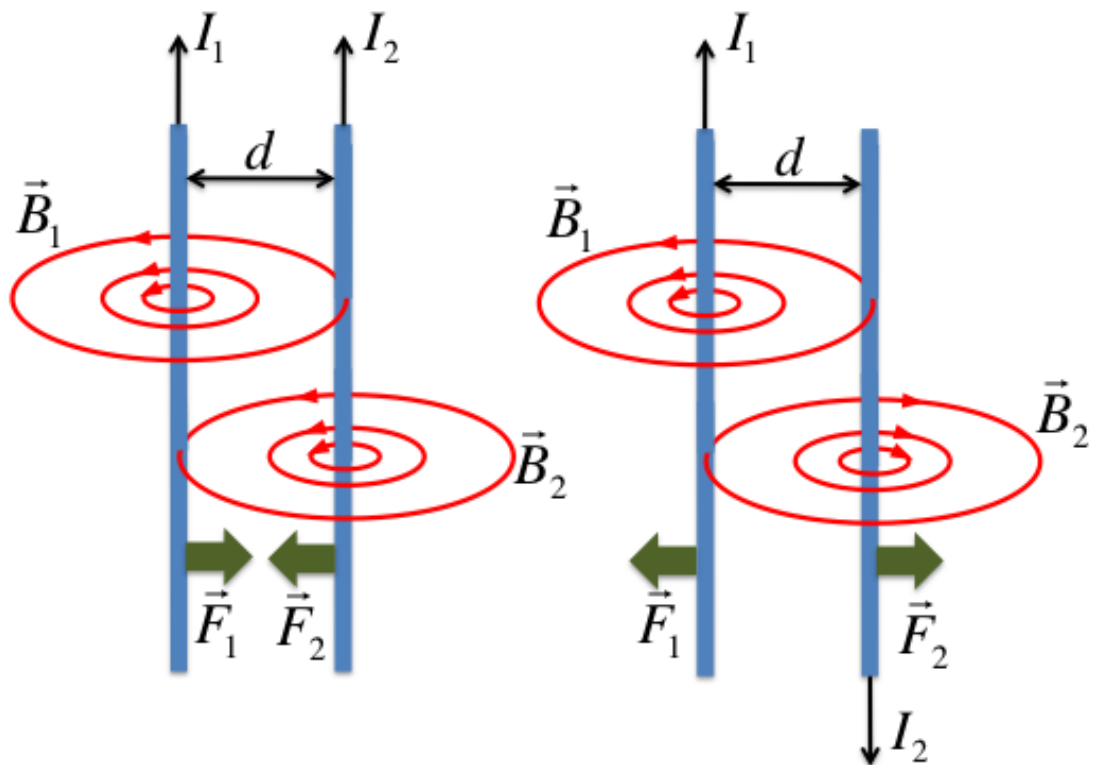
In 1792 **Volta** invented the “pila”.



Alessandro Volta
(Italy, 1745-1827)

Ampère's Law

On 1820 **Ampère** demonstrated that parallel wires carrying currents attract or repel each other. This laid the foundation of **electrodynamics**.



André-Marie Ampère
(France, 1775-1836)

(in nowadays notation and International System (SI) Units)

Ohm's Law

In 1827 **Ohm** published the book "*Die galvanische Kette, mathematisch bearbeitet*" (The Galvanic Circuit Investigated Mathematically) where his law is formulated.

$$S = \gamma E$$

current flowing in a wire

difference of "electroscopic force" at wire terminals

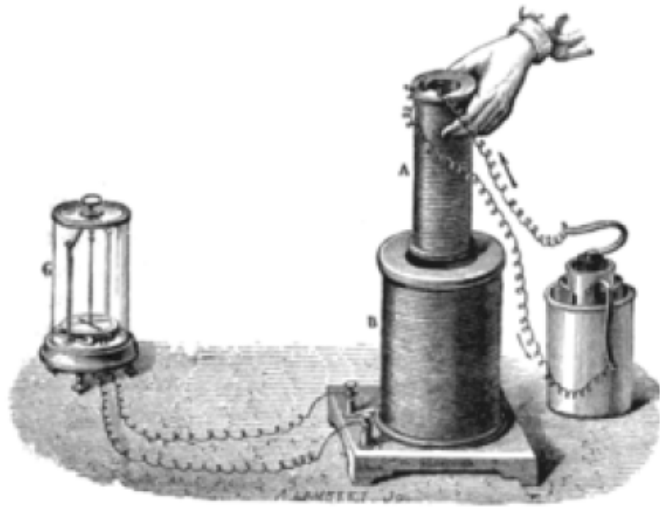
Ohm's law connected the theory of currents with the older theory of electrostatic.



George Simon Ohm
(Germany, 1789-1854)

Electromagnetic Induction

In 1831 **Faraday** discovered the electromagnetic induction.



Michael Faraday
(England, 1791-1867)

“Distinct conversion of Magnetism into Electricity” (Faraday)

Electrotonic state

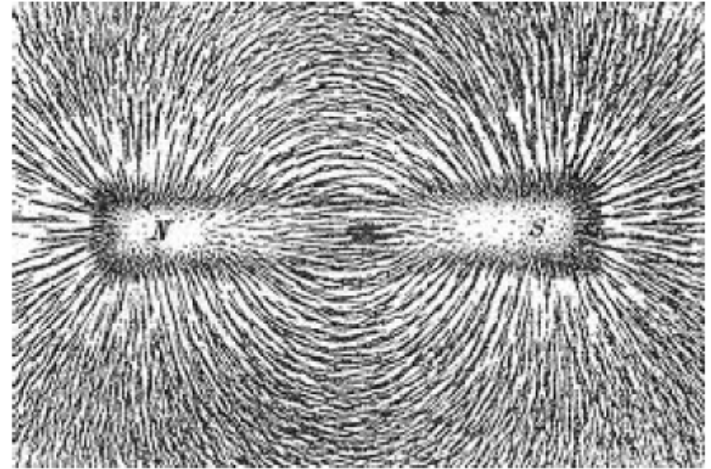
“This state does not manifest itself by any known phenomena as long as it is undisturbed, but any change in this state is indicated by a current or tendency towards a current.” (J. C. Maxwell)

Faraday's Law

Faraday determined the law of induction of currents by introducing a new way of representing the magnetic field.

Lines of magnetic force

“By magnetic curves, I mean the lines of magnetic forces, however modified by the juxtaposition of poles, which could be depicted by iron fillings; or those to which a very small magnetic needle would form a tangent.” (Faraday, Exp. Res. § 114)



With these lines Faraday conceived all space to be filled.

“whether the wire moves directly or obliquely across the lines of force, in one direction or another, it sums up the amount of the forces represented by the lines it has crossed”, so that “the quantity of electricity thrown into a current is directly as the number of curves intersected”. (Faraday, Exp. Res. § 114)

The induced electromotive force is proportional to the number of the unit lines of magnetic force intersected by the wire per second.

Lenz's Law

In 1834 **Lenz** formulated his law which can be enunciated as it follows:

“when a conducting circuit is moved in a magnetic field, the induced current flows in such a direction that the ponderomotive forces on it tend to oppose the motion”.



Heinrich Friedrich Emil Lenz
(Baltic German, 1804-1865)

Gauss's Intuitions

In 1835 **Gauss** suggested that electromagnetic theory should be derived from a single equation for the force between moving charges, which should be function of the relative velocity of the charges besides the distance between them.



Karl Friederich Gauss
(Germany, 1777-1855)

In a letter to Weber (on March 19, 1845) Gauss remarked that this force should also include retardation:

the “electric action” should propagate between the charges with a finite velocity.

“On the Motion of Electricity in Wires”

In 1855 **Kelvin** investigated the transients in long cables assuming that the magnetic effects were negligible. The work of Kelvin was followed by a celebrated paper of Kirchhoff.

In 1857 **Kirchhoff** published “*On the Motion of Electricity in Wires*”.

THE
LONDON, EDINBURGH AND DUBLIN
PHILOSOPHICAL MAGAZINE
AND
JOURNAL OF SCIENCE.

[FOURTH SERIES.]

JUNE 1857.

LIV. *On the Motion of Electricity in Wires.*
By G. KIRCHHOFF*.

I HAVE attempted to establish a general theory of the motion of electricity in an infinitely thin wire, by assuming certain facts which are observed in constant currents, and in currents whose intensity alters but slowly, to be universally valid. I will here develop this theory, and show its application to some cases of a simple nature.



William Thomson
(Lord Kelvin)
(England, 1824-1907)



Gustav Robert
Kirchhoff
(Germany, 1824-1887)

“On the Motion of Electricity in Wires”

By using Weber's electrodynamics and the charge conservation law Kirchhoff foresaw before Maxwell that an electric wave propagates along a conducting wire with the velocity of light in vacuum.

...

The velocity of propagation of an electric wave is here found to be $= c$ hence it is independent of the cross section, of the conductivity of the wire, also, finally, of the density of the electricity: its value is 41950 German miles in a second, hence very nearly equal to the velocity of light *in vacuo*.

...

Maxwell's Equations

- ◆ In 1855 and 1856 **Maxwell** published "*On Faraday's lines of force*"

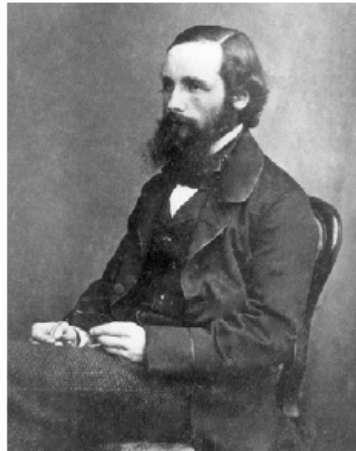
Transactions of the Cambridge Philosophical Society, Vol. X, Part I 1855; Part II 1856.

- ◆ In 1861 and 1862 **Maxwell** published "*On Physical lines of force*"

Phil. Mag., XXI, pp. 161-175, 281-291, 338-348, 1861 (Parts I and II); Phil. Mag., XXIII, pp. 12-25, 85-95, 1862 (Parts III and IV).

- ◆ In 1865 **Maxwell** published "*A Dynamical Theory of the Electromagnetic Field*"

Phil. Trans. Roy. Soc., vol. 155, 1865, pp. 459-512.

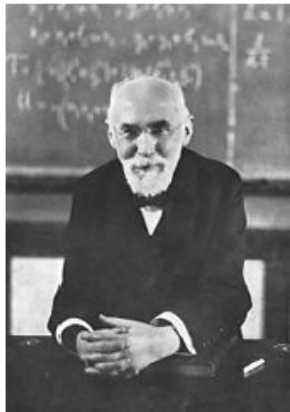


James Clerk Maxwell
(Scotland, 1831-1879)

Lorentz's Force Law

In 1895 **Lorentz** introduced his force law.

The Maxwell's equations do not furnish the force exerted on the charges and currents, they only give the electromagnetic field generated by them.



Hendrik Antoon Lorentz
(Honland, 1853-1928)

The **Lorentz's force** combined to **Newton's equation** tell us how the charges react in presence of an electromagnetic field.

Retarded Electromagnetic Potentials

In 1867 **Lorenz** introduced the retarded electromagnetic potentials. By generalizing the Riemann's intuition, Lorenz proposed that the magnetic vector potential were retarded, too.

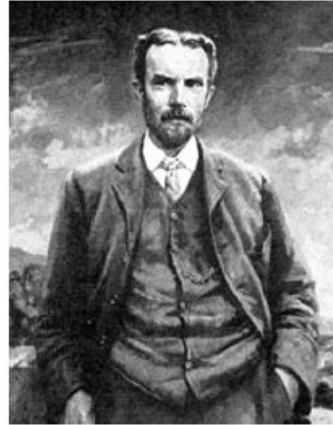


Lorenz

Ludwig Lorenz
(Denmark, 1829-1891)

Maxwell's Equations in Vector Form

The Maxwell's equations in the nowadays vector notation were formulated by **Heaviside** in 1886.



Oliver Heaviside
(England, 1850-1925)

Hertz's Theory

In 1884 **Hertz**, Helmholtz's pupil, published a paper in which starting from the older action-at-distance theory (Weber's electrodynamics) proceeded to obtain Maxwell's equations in an alternate way.



Heinrich Rudolf Hertz
(Germany, 1857-1894)



Hermann Ludwig Ferdinand
von Helmholtz
(Germany, 1857-1894)

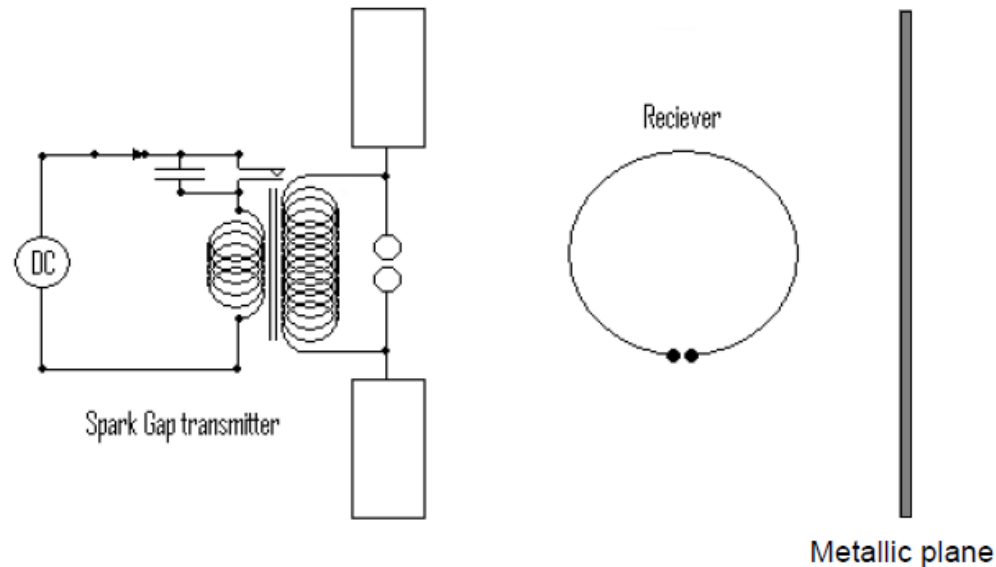
Hertz avoided both the mechanical models of Maxwell and his formulation of displacement current by using successive approximation from the earlier action-at-distance theory together with a principle of the “unity of the electric force”.

However, Hertz's conclusions were: “If the only choice lay between the usual system and Maxwell's, than the latter has undoubtedly the advantage”.

This results has probably prompted Hertz to search vigorously for the experimental evidence of Maxwell's electromagnetic theory.

Hertz's Experiment on Radio Waves

In 1888 **Hertz** confirmed experimentally the existence of electromagnetic waves, other than light (at radiofrequency) and so validated the Maxwell's Equations.



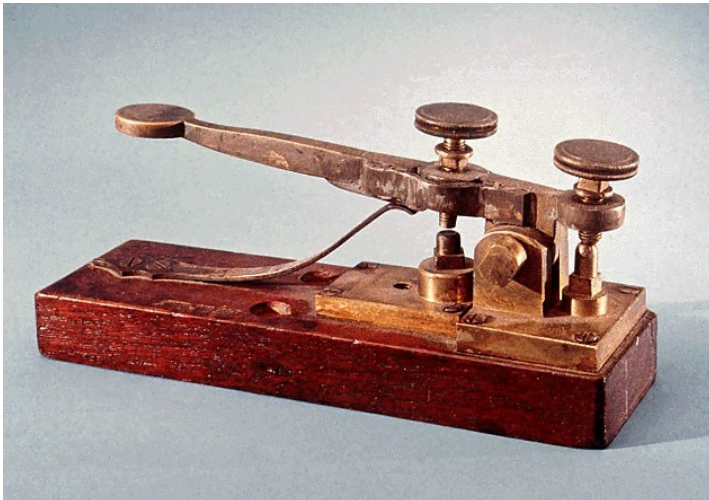
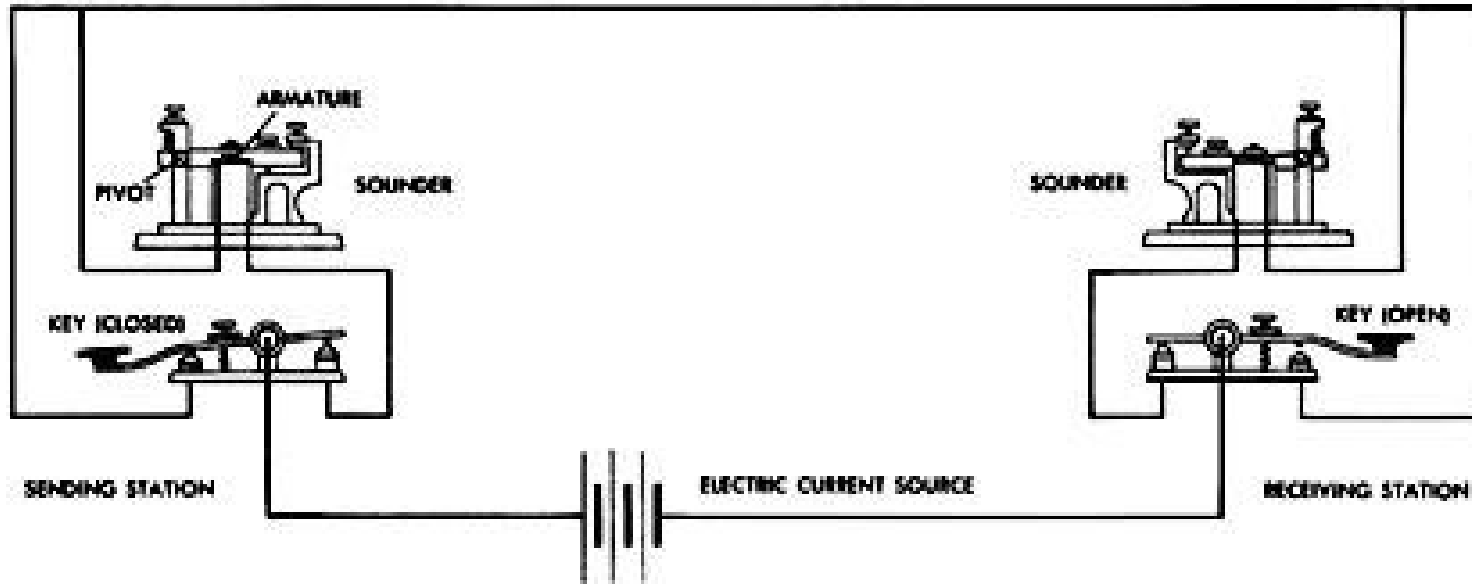
In contrast to the 1884 paper, in his 1890 paper "*On the fundamental Equations of Electromagnetics for bodies at Rest*" Hertz postulated Maxwell's equations, rather than deriving them in an alternative form.

Cooke and Wheatstone's five-needle, six-wire telegraph

The first commercial electrical telegraph, the Cooke and Wheatstone telegraph, was co-developed by William Fothergill Cooke and Charles Wheatstone. In May 1837 they patented a telegraph system which used a number of needles on a board that could be moved to point to letters of the alphabet. The patent recommended a five-needle system, but any number of needles could be used depending on the number of characters it was required to code. A four-needle system was installed between Euston and Camden Town in London on a rail line being constructed by Robert Stephenson between London and Birmingham. It was successfully demonstrated on 25 July 1837



It is started with Telegraph

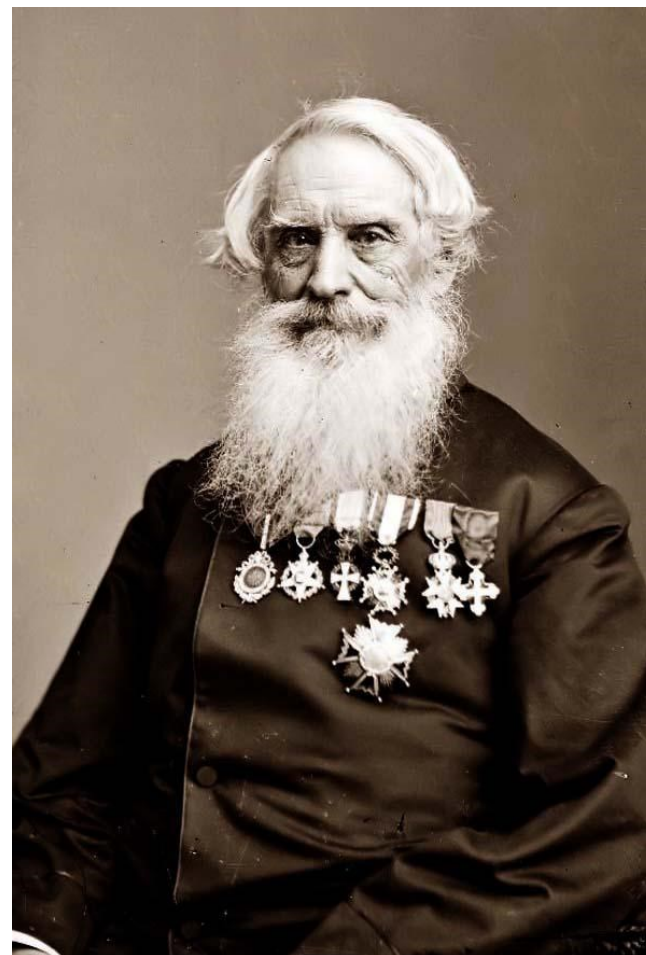


- Used a an on-off switch – the telegraph key – to turn an electric current on and off, sending pulses of current through a wire to an electromagnet that would click in time to the pulses

Morse Code Alphabet

The International morse code characters:

A .-	N -.	0 -----
B -...	O ---	1 .----
C -.-.	P .-.-	2 ..---
D -..	Q --.-	3 ...--
E .	R .-.	4-
F ..-.	S ...	5
G --.	T -	6 -....
H	U ..-	7 --...
I ..	V ...-	8 ---..
J .----	W .--	9 ----.
K -.-	X -.-	Fullstop .----
L .-..	Y -.-	Comma --.---
M --	Z --..	Query ..----



Portrait of Samuel F. B. Morse taken by Mathew Brady, in 1866. Medals worn (from wearer's right to left, top row: Nichan Iftikhar (Ottoman); Order of the Tower and Sword (Portugal); Order of the Dannebrog (Denmark); Gold Medal of Art and Science (Württemberg); Legion of Honour (France); Order of Saints Maurice and Lazarus (Italy). Bottom row: Order of Isabella the Catholic (Spain)

Novelty Stage

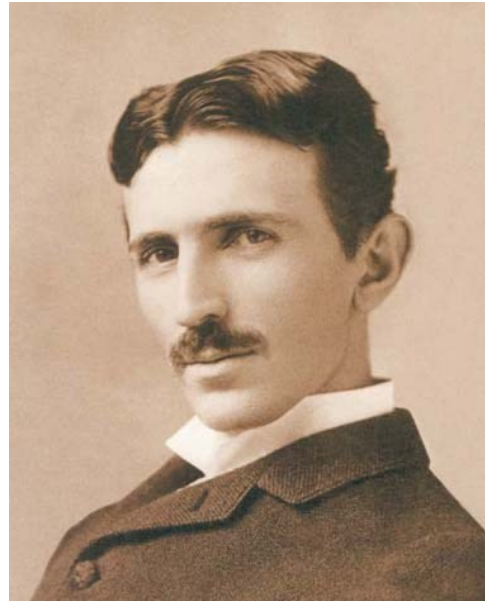
THREE GUYS WHO INVENTED THE DEVICE TO TRANSMIT A SIGNAL THROUGH THE AIR (1894)

A. Guglielmo Marconi vs.

B. NiKola Tesla vs.

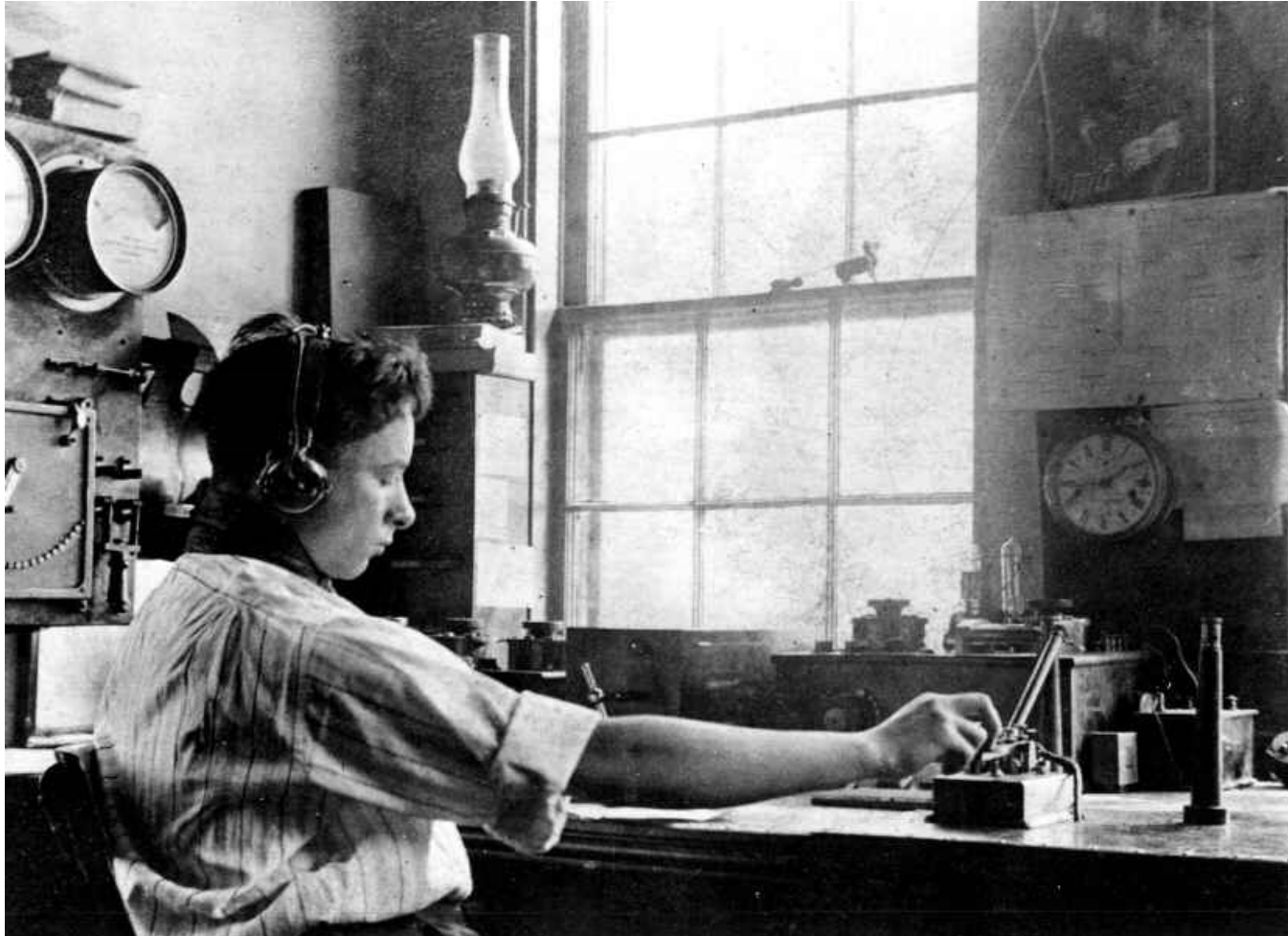
C. Alexander Popov

– Who invented Wireless telegraphy?



Entrepreneurial Stage

- David Sarnoff and the Titanic

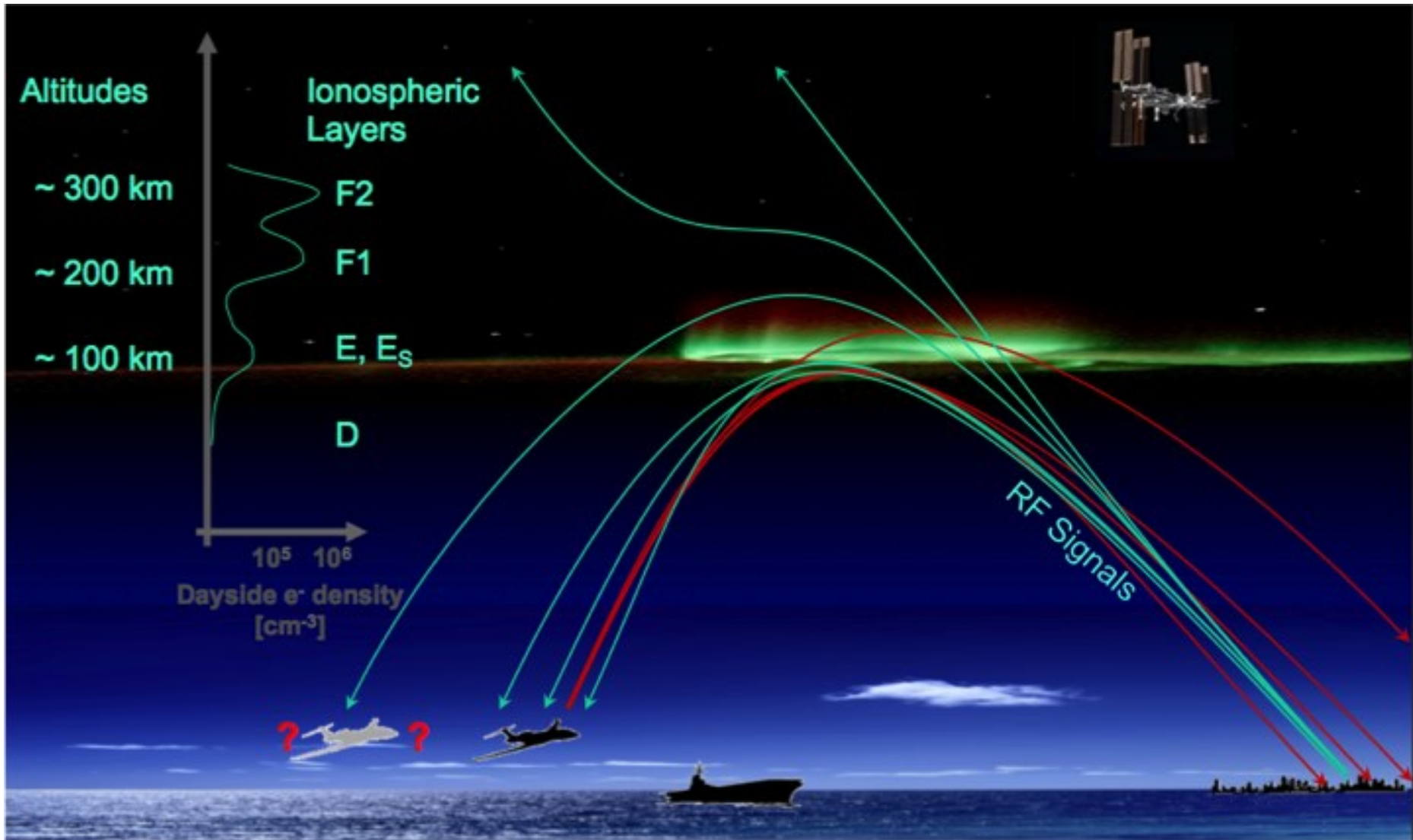


Still using Morse code with radio

Radio Frequency Spectrum: Ranges

Designation	Abbreviation	Frequencies	Wavelengths
Very Low Frequency	VLF	3 kHz - 30 kHz	100 km - 10 km
Low Frequency	LF	30 kHz - 300 kHz	10 km - 1 km
Medium Frequency	MF	300 kHz - 3 MHz	1 km - 100 m
High Frequency	HF	3 MHz - 30 MHz	100 m - 10 m
Very High Frequency	VHF	30 MHz - 300 MHz	10 m - 1 m
Ultra High Frequency	UHF	300 MHz - 3 GHz	1 m - 100 mm
Super High Frequency	SHF	3 GHz - 30 GHz	100 mm - 10 mm
Extremely High Frequency	EHF	30 GHz - 300 GHz	10 mm - 1 mm

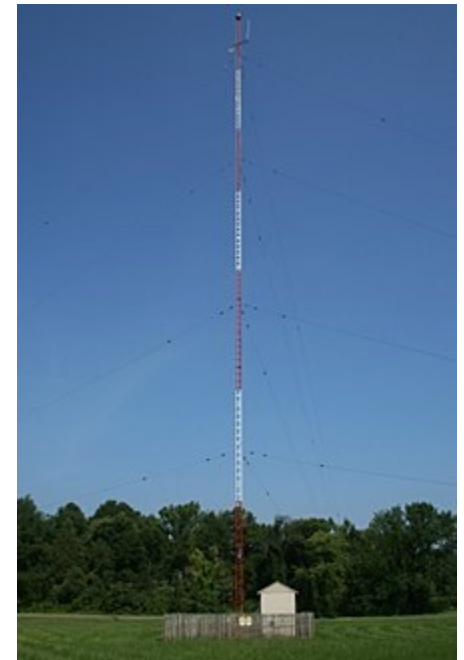
Ionosphere



Medium frequency (MF)

300 kHz to 3 MHz 1,000 m to 100 m

- **AM radio stations**



Mast radiator of a commercial MF AM broadcasting station, Chapel Hill, North Carolina, USA

In telephony, the usable voice frequency band ranges from approximately 300 Hz to 3400 Hz. It is for this reason that the ultra low frequency band of the electromagnetic spectrum between 300 and 3000 Hz is also referred to as voice frequency, being the electromagnetic energy that represents acoustic energy at baseband.

VLF Communication



Satellite Communication

